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基於自我確信評估與問題拆解的自適應知識檢索方法

SEEK: Self-certainty Evaluated Evidence-gated Knowledge-retrieval

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學號 Student ID: S113065528

研究生 Author： 簡宏諭 Hung-Yu Chien

指導教授 Advisor： 高宏宇 博士 Dr. Hong-Yu Kao

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Abstract

This study proposes SEEK (Self-certainty Evaluated Evidence-gated Knowledge-retrieval), a lightweight adaptive Retrieval-Augmented Generation (RAG) framework for open-domain question answering. Standard RAG systems apply a fixed retrieval strategy to every query, which can inject noise on simple questions and miss intermediate evidence on multi-hop questions. Existing adaptive methods either route at a coarse question level, probe uncertainty at a noisy token level, or learn expensive sub-question policies through tree search and preference optimisation.

SEEK uses a compact KG-triple decomposer trained with a single LoRA SFT pass on a 12K mixture of MuSiQue, 2WikiMultihopQA, and PopQA examples. The final system no longer treats model confidence as a single hard skip gate. Instead, it generates both a direct answer and a decomposed answer, gates sub-question retrieval with tail-leakage surprisal, preserves original-question evidence inside decomposed sub-question contexts, and applies a training-free verifier with a conservative type-aware fallback. This late-routing design preserves the interpretability of confidence-based methods while avoiding the brittleness of a global threshold.

Under the honest wiki-21M retrieval protocol aligned with DeepRAG, SEEK reaches an average of 41.62 EM and 48.66 F1, exceeding DeepRAG's 38.45 EM and 45.75 F1. SEEK improves over DeepRAG on 2WikiMultihopQA (48.70 EM), MuSiQue (26.60 EM), and PopQA (53.70 EM), while remaining behind on HotpotQA (37.50 EM versus 40.70 EM). Analysis shows that direct and decomposed paths are complementary, that oracle direct-vs-decompose routing still offers 3.8 to 6.1 EM of headroom, and that tail-leakage surprisal is useful as a sub-question retrieval signal but not as a correctness guarantee.

Keywords: Retrieval-Augmented Generation, Adaptive Retrieval, Multi-hop Question Answering, Tail-Leakage Surprisal, Verifier Routing, Large Language Models

摘要

本研究提出 SEEK (Self-certainty Evaluated Evidence-gated Knowledge-retrieval)，一個用於開放領域問答的輕量化自適應檢索增強生成 (Retrieval-Augmented Generation, RAG) 框架。傳統 RAG 對所有問題採用固定檢索策略：簡單問題可能因不必要的檢索引入雜訊，多跳問題則可能因單次檢索無法找到中介證據而回答失敗。既有自適應方法不是在問題層級做過於粗糙的路由，就是在詞元層級使用嘈雜的不確定性訊號，或像 DeepRAG 一樣透過樹搜尋與偏好最佳化學習昂貴的子問題決策策略。

SEEK 使用一個以知識圖譜三元組格式表示的分解器，在 MuSiQue、2WikiMultihopQA 與 PopQA 混合而成的一萬兩千筆資料上，以 LoRA 進行單次監督式微調。最終版本不再把單一信心訊號當成硬性的跳過檢索門檻，而是同時產生直接作答與分解作答兩個答案，以尾端洩漏驚異度控制子問題檢索，並將原始問題檢索到的證據加回子問題的作答上下文，最後由無需額外訓練的驗證器搭配依答案類型設計的保守回退機制選出最終答案。這種延後路由的設計保留了信心訊號的可解釋性，同時避免單一全域閾值在跨資料集時的不穩定。

在與 DeepRAG 對齊、以兩千一百萬篇維基百科段落為檢索語料的設定下，SEEK 平均達到 41.62 EM 與 48.66 F1，高於 DeepRAG 的 38.45 EM 與 45.75 F1。分資料集來看，SEEK 在 2WikiMultihopQA 達到 48.70 EM、MuSiQue 達到 26.60 EM、PopQA 達到 53.70 EM，皆高於 DeepRAG；HotpotQA 則為 37.50 EM，仍低於 DeepRAG 的 40.70 EM。分析顯示直接路徑與分解路徑具有互補性，若能在兩條路徑之間做出理想的選擇，仍有 3.8 到 6.1 EM 的提升空間；尾端洩漏驚異度適合作為子問題檢索的控制訊號，但不能視為答案正確與否的保證。

關鍵字：檢索增強生成、自適應檢索、多跳問答、尾端洩漏驚異度、驗證器路由、大型語言模型

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Chapter 1 Introduction

1.1 Background and Motivation

Large language models (LLMs) often hallucinate when queried about recent, obscure, or proprietary information outside their training scope. Retrieval-Augmented Generation (RAG) addresses this by fetching relevant external evidence, reducing factually unsupported answers [1], [2]. However, applying RAG indiscriminately creates two problems. For simple questions, unnecessary retrieval can inject distracting evidence and increase latency. For complex multi-hop questions, a single retrieval step often fails to surface the intermediate entities needed for reasoning.

Adaptive RAG therefore needs to decide not only *whether* retrieval is useful, but also *when to trust decomposition*. Decomposition can sharpen retrieval queries, but it can also remove the original question's comparison targets, entity pairs, or disambiguating clues. This thesis frames adaptive RAG as a *decomposition-routing problem*: the system should control retrieval at the sub-question level while preserving the original-question path as a competing candidate.

1.2 Research Objectives

This thesis introduces SEEK (Self-certainty Evaluated Evidence-gated Knowledge-retrieval), a lightweight adaptive RAG framework built around training-free verifier routing. The objectives are:

- Build a compact KG-triple decomposer that converts multi-hop questions into atomic `entity >> relation` sub-questions.
- Use token-level tail-leakage surprisal as an interpretable confidence signal, while showing that confidence alone is insufficient as a global hard router.
- Generate both direct and decomposed candidate answers, preserve original-question evidence inside decomposed steps, and select the final answer with a training-free

verifier based on confidence, answer type, decomposition stability, and candidate agreement.

- Evaluate the final system under the honest wiki-21M retrieval protocol used by DeepRAG, showing where a lightweight verifier can match or exceed a learned retrieval policy and where routing headroom remains.

Chapter 2 Related Work

2.1 Adaptive RAG

Retrieval-Augmented Generation (RAG) [1] augments a parametric language model with a non-parametric retriever, allowing the model to condition its generation on external evidence rather than relying only on parameters learned during pretraining. A typical RAG pipeline can be described as five stages [3]:

1. **Indexing.** The source corpus is segmented into passages, and each passage is stored in a form that supports retrieval, such as sparse lexical indexes or dense vector representations.

2. **Query manipulation.** The input question may be rewritten, expanded, normalised, or decomposed so that the retriever can better match it with relevant evidence.

3. **Retrieval.** The system selects candidate passages from the indexed corpus, commonly using sparse lexical matching, dense vector similarity, or a hybrid retrieval strategy.

4. **Post-retrieval processing.** The initially retrieved passages may be filtered, reranked, compressed, or otherwise refined before they are passed to the generator.

5. **Generation.** The language model conditions on the selected evidence and produces the final answer, ideally grounding its output in the retrieved passages.

This retrieve-then-generate paradigm is now widely used to make language model outputs more factual in question answering, explanation writing, and other knowledge-intensive tasks. Its central advantage is that factual evidence can be updated externally without retraining the language model. For multi-hop question answering, however, the quality of the retrieved evidence depends strongly on the query that is issued at each step: a single original question may not contain the intermediate entity needed to retrieve the next supporting passage.

This fixed retrieval assumption creates two well-documented failure modes:

- **Noise injection on easy queries.** For questions the model already knows (e.g. "Who wrote *Hamlet*?"), retrieved passages introduce distractor entities that can override correct parametric knowledge [4], [5].
- **Insufficient evidence on hard queries.** For multi-hop questions, a single retrieval pass rarely surfaces all bridging entities, leading to incomplete or compositionally wrong answers [6].

These two failure modes motivate adaptive retrieval: deciding *when* retrieval is useful, *how much* evidence to fetch, and what signal should determine the gate.

Rather than assuming a fixed retrieve-then-read pipeline, adaptive RAG changes the retrieval path according to the input question, the model state, or the current reasoning step. Some methods decide before generation, using a complexity router or a self-knowledge estimate. Others trigger retrieval from uncertainty observed during generation. A third group learns retrieval actions as part of the generation or reasoning policy. These distinctions structure the discussion below.

Question-level routing. A first line of work makes the retrieval decision before generation. These systems inspect the input question and route it to a direct-answer path or a retrieval pipeline, often using a trained classifier or a self-knowledge estimate.

Adaptive-RAG [7] routes each incoming query to one of three retrieval strategies according to its predicted complexity. Easy questions are answered without retrieval, medium-complexity questions use a single retrieval step, and complex questions are routed to a multi-step retrieve-and-reason pipeline such as IRCoT [6]. The router is a T5-Large classifier trained with automatically derived labels. Each training question is assigned to the cheapest strategy that still produces the correct answer, so the classifier learns a cost-aware mapping from question complexity to retrieval depth. This design reduces unnecessary retrieval compared with always using a multi-step pipeline, but the route itself remains a coarse question-level decision.

SKR [8] approaches adaptive retrieval from the perspective of self-knowledge. Instead of predicting general question complexity, SKR estimates whether the language model is likely to know the answer from its own parametric memory. If the question

is judged as known, the model answers directly; if it is judged as unknown, external retrieval is invoked. This judgement can be implemented with a trained classifier or with memory-based self-knowledge strategies. In the classifier setting, SKR is a binary retrieval gate: it decides whether to retrieve, but does not distinguish between single-step and multi-step retrieval. Therefore, compared with Adaptive-RAG, SKR gives a more direct estimate of whether retrieval is needed, but it still commits to a question-level decision before reasoning begins.

Uncertainty-triggered retrieval. A second line of work delays the retrieval decision until generation, using uncertainty signals from the model itself. These methods are often training-free, since the retrieval decision is derived from token probabilities, entropy, attention patterns, or internal model states rather than from an additional trained classifier.

FLARE [9] moves the retrieval decision from the question level to the generation level. At each step, it first drafts a tentative next sentence and checks whether the sentence contains low-confidence tokens. If the confidence falls below a threshold, the tentative sentence is used to formulate a retrieval query, and the model regenerates the sentence with the retrieved evidence. This design allows retrieval to occur only when the model appears uncertain during generation. However, token probability is only an indirect proxy for factual uncertainty. A low-probability token may reflect lexical variation rather than missing knowledge, and sentence-level regeneration may not align well with multi-hop reasoning, where the natural decision unit is often a sub-question.

DRAGIN [10] further refines dynamic retrieval by considering both when to retrieve and what to retrieve. Its *Real-time Information Needs Detection* (RIND) module estimates whether the current generation state requires external evidence, while its *Query Formulation based on Self-attention* (QFS) module constructs retrieval queries from salient context tokens instead of relying only on the most recent sentence. Compared with FLARE, DRAGIN provides a more context-aware retrieval trigger and query formulation strategy. Nevertheless, its decision is still made at the token-generation level, which can be mismatched with explicit sub-question reasoning, and it introduces additional overhead from attention-based query construction.

SeaKR [11] estimates retrieval need from the model's internal states rather than from

output probabilities alone. It activates retrieval when the model exhibits high self-aware uncertainty and reranks retrieved passages according to how much they reduce that uncertainty. This makes the retrieval signal more closely tied to the model's internal generation dynamics than token-level probability alone. However, the signal is still an estimate of uncertainty rather than a direct measure of answer correctness. A model may be uncertain for reasons unrelated to missing external knowledge, and retrieval may not necessarily improve the answer even when uncertainty is high.

TARG [12] studies adaptive retrieval as a training-free, single-shot decision. It generates a short no-context draft and computes uncertainty from the prefix logits to decide whether retrieval should be invoked. In addition to entropy, TARG considers a top-1/top-2 logit-margin signal and a small-sample variance signal. Its main strength is that it provides a lightweight retrieval gate without training an auxiliary classifier or modifying the language model. However, TARG makes one gating decision for the whole question, even on multi-hop benchmarks such as MuSiQue. Its behaviour in decomposition-based retrieval settings, where each sub-question may need its own decision, therefore remains less explored.

Across this family, uncertainty provides a useful and interpretable signal for adaptive retrieval, but it remains an imperfect proxy. Low uncertainty does not guarantee a correct answer, and high uncertainty does not always indicate that retrieval will help. SEEK addresses this limitation by using tail-leakage surprisal both as a sub-question retrieval signal and as the confidence component of a late verifier, rather than treating a single uncertainty score as a hard router for the entire question.

Learned retrieval policies. A third line of work makes retrieval control part of the model's learned generation or reasoning behaviour, rather than relying on a front-end router or a hand-set uncertainty threshold.

Self-RAG [13] trains the language model to generate special reflection tokens that control retrieval and evaluate retrieved passages. For example, the model can emit `[Retrieve]` to decide whether external evidence is needed, and then use additional tokens to judge whether a passage is relevant, supportive, or useful. This design makes retrieval part of generation itself, rather than a separate preprocessing step. However, it requires fine-

tuning with reflection-token supervision and a custom decoding procedure that interleaves normal text generation with retrieval-related decisions.

DeepRAG [14] formulates retrieval-augmented reasoning as a step-by-step decision process. At each sub-query, the model chooses between answering from parametric memory and issuing a retrieval call. Its training procedure first uses binary tree search to explore both choices for each sub-query, and then applies preference-based calibration to refine the retrieval decision at each sub-query. This design gives it strong sub-question-level retrieval control, but the cost is substantial. Tree-search rollout becomes expensive as the number of reasoning steps grows, preference pairs must be carefully constructed, and the final retrieval policy is learned implicitly rather than exposed as an interpretable score.

A related line of work studies retrieval cost more directly. **Fast-or-Better** [15] trains two complexity classifiers, one prioritising accuracy and the other prioritising retrieval efficiency, and exposes a user-controlled parameter that interpolates between them, so the accuracy–cost trade-off is no longer fixed at a single operating point. This view reinforces an important point for adaptive RAG: retrieval should be evaluated not only by accuracy, but also by the cost required to obtain that accuracy.

SEEK is closest in spirit to DeepRAG because both methods make retrieval decisions at the sub-question level. The key difference is that SEEK avoids tree search and preference optimisation. It uses a single supervised decomposer and replaces the learned retrieval policy with a training-free verifier based on confidence, decomposition stability, answer type, and candidate agreement.

Overall, SEEK occupies the under-explored region of *late verifier-based routing with interpretable, training-free signals*: only a single SFT pass is needed for the decomposer, and the final route is decided by transparent query-level diagnostics rather than a learned policy.

2.2 Reasoning and Knowledge Boundaries in RAG

A parallel thread focuses on *how* to retrieve and reason once retrieval is deemed necessary, independent of the gate.

The foundational technique is **IRCoT** [6], which interleaves chain-of-thought rea-

soning with retrieval: after each generated CoT sentence, the partial reasoning becomes the next query. IRCoT is the canonical "multi-step" retrieve-and-reason implementation, but its retrieval count grows linearly in the CoT length, regardless of whether intermediate steps are factually difficult. Question decomposition provides a complementary way to structure multi-hop QA. Rather than retrieving after each free-form reasoning sentence, decomposition-based methods explicitly split a complex question into simpler sub-questions that can be answered independently or sequentially. Early systems such as **DecompRC** [16] show that compositional questions can be reduced to single-hop reading-comprehension steps, while **BREAK/QDMR** [17] formalises decomposition as an ordered sequence of natural-language operations required to answer the original question. SEEK inherits this decomposition paradigm, but uses it primarily as an inspectable reasoning path: it preserves the original-question context for decomposed steps, evaluates whether the generated sub-questions are useful, and verifies whether the direct or decomposed answer should be trusted.

More recent work synthesises reasoning data to internalise the retrieve-and-reason loop. **Auto-RAG** [18] autonomously synthesises multi-turn retrieval–reasoning trajectories and fine-tunes on them, so that the model plans its own retrieval steps without hand-crafted prompts. **Search-o1** [19] embeds retrieval inside a large reasoning model in an agentic loop, augmenting o1-style long-chain reasoning with on-demand search and a document-reasoning module that distils retrieved passages before they re-enter the chain. **AirRAG** [20] activates the model's intrinsic reasoning by combining Monte Carlo Tree Search with self-consistency to explore alternative solution paths. These methods substantially strengthen the reasoning side of RAG, but they rely on heavy data synthesis or search-time expansion; SEEK instead keeps the reasoning component a single lightweight SFT decomposer and concentrates its novelty in training-free verification.

Every adaptive method above ultimately rests on a model's ability to recognise its own *knowledge boundary* — the line between what it reliably knows and what it must look up. A dedicated line of work studies this capacity directly, and we group it into three strands: *self-knowledge elicitation*, *uncertainty estimation*, and the *interaction between the knowledge boundary and retrieval*.

Self-knowledge and its calibration. A first strand asks whether a model can introspect on what it knows. Kadavath et al. [21] show that LLMs are, to a useful degree, calibrated: a model can be prompted to predict the probability `P(True)` that its own answer is correct, and these self-judgements correlate with actual accuracy — though calibration degrades on out-of-distribution questions. Yin et al. [22] construct a "SelfAware" benchmark of answerable and unanswerable questions and find that LLMs only partially "know what they don't know": self-knowledge can be elicited but lags well behind human performance. From the supervised side, Kapoor et al. [23] argue that models "must be taught to know what they don't know", demonstrating that lightweight calibration-tuning or probing substantially sharpens a model's estimate of its own uncertainty. Yin et al. [24] formalise the notion of a knowledge boundary itself, proposing to benchmark a model by probing the frontier of its reliable knowledge rather than by aggregate accuracy alone.

Uncertainty estimation as a boundary signal. A second strand asks how to *measure* the boundary at inference time. Beyond retrieval-specific methods, recent work on LLM uncertainty quantification compares several confidence estimators for answer correctness, including token entropy, maximum softmax probability, margin-based scores, verbalised confidence, and response consistency [25]. These comparisons show that the reliability of each signal varies across tasks and models. This motivates testing whether entropy is sufficient in SEEK's QA setting, or whether a signal focused on the probability mass escaping the top candidates provides a more discriminative retrieval gate.

Semantic entropy further shows why token-level uncertainty can be limited for natural-language answers: different surface forms may express the same meaning, so uncertainty over token sequences does not always correspond to uncertainty over answers. Kuhn et al. [26] introduce *semantic entropy*, which clusters multiple sampled answers by meaning before computing entropy, so that paraphrases of one correct answer do not register as uncertainty; Farquhar et al. [27] scale this idea and show that semantic entropy reliably detects confabulations across QA and biography generation. This observation is important for SEEK because exact QA answers often involve aliases, entity variants, and short spans. The same concern shapes SEEK's choice of signal: competition among the top candidates is often alias-level variation, so tail-leakage surprisal counts only the probabil-

ity mass that escapes the candidate set. We nevertheless treat token-level confidence as a cheap diagnostic signal rather than as a correctness guarantee.

The boundary under retrieval. A third strand studies the boundary specifically in the RAG setting. Ren et al. [5] show that retrieval augmentation not only injects evidence but also *reshapes* the boundary: relevant passages let a model answer questions it could not before, while irrelevant or distracting passages can push a previously-correct answer over the boundary into error — precisely the "noise injection on easy queries" failure mode of §2.1. This makes a per-step boundary estimate, rather than a one-shot query-level one, essential when retrieval and reasoning are interleaved.

These three strands form the conceptual foundation of SEEK. The failure modes of §2.1 — spurious retrieval on easy queries, missing retrieval on hard ones, and confident hallucination under a hard confidence threshold — are all symptoms of a poorly localised knowledge boundary. Where confidence-based methods read a single uncertainty proxy and policy-based methods bury the boundary inside a learned policy, SEEK makes the boundary inspectable through a late verifier: confidence, decomposition stability, and answer agreement jointly decide whether the direct or decomposed path should be trusted.

2.3 Evaluation Datasets

The evaluation of adaptive RAG hinges on datasets that span both single-hop and multi-hop complexity. This thesis uses HotpotQA, 2WikiMultihopQA, MuSiQue, and PopQA to cover both multi-hop reasoning and single-hop factual recall.

HotpotQA. HotpotQA [28] is the canonical multi-hop QA benchmark, containing 113K Wikipedia-based questions requiring 2-hop reasoning. Each example provides (a) the question, (b) the gold answer, (c) two gold-supporting paragraphs mixed with eight distractor paragraphs (in the `distractor` setting), and (d) sentence-level supporting facts. Questions fall into two compositional patterns: *bridge* (the answer to sub-question 1 is the subject of sub-question 2) and *comparison* (two entities are compared along a shared attribute). It is a standard evaluation set for adaptive RAG and multi-hop reasoning systems.

2WikiMultihopQA. 2WikiMultihopQA [29] constructs 192K multi-hop questions by leveraging Wikidata knowledge-graph paths, ensuring each question requires traversal of an explicit reasoning chain. It includes four question types (comparison, inference, compositional, and bridge-comparison) and provides KG-triple annotations alongside textual evidence. Compared to HotpotQA, 2WikiMultihopQA is constructed so that its questions are hard to answer through reasoning shortcuts, making it a more reliable test of compositional reasoning.

MuSiQue. MuSiQue (**Mu**ltihop questions via **Si**ngle-hop **Que**stions) [30] addresses two pathologies of earlier datasets: (i) shortcut answers via single-hop reasoning, and (ii) weak distractor selection. MuSiQue is built by composing 2–4 single-hop questions into chains using strict connectivity criteria, yielding 25K questions of 2-hop, 3-hop, and 4-hop varieties. Importantly, MuSiQue provides the *decomposed* sub-questions as supervision, which makes it a natural training source for decomposer models such as the one used in SEEK. Its high distractor difficulty makes it the hardest of the four datasets in this thesis.

PopQA. PopQA [4] is a *single-hop* entity-centric QA dataset of 14K questions, each derived from a Wikidata subject–property–object triple. Critically, PopQA tags each entity with its *popularity* (measured by monthly Wikipedia page views), enabling stratified evaluation of factual recall on long-tail entities. Despite being single-hop, PopQA exposes the over-decomposition failure mode of decomposer-based pipelines: trained on multi-hop data, a decomposer may emit two-step plans even when one step suffices, harming both EM and efficiency.

Single-hop open-domain QA. For completeness, prior adaptive-RAG papers also evaluate on **Natural Questions (NQ)** [31], **TriviaQA** [32], and **SQuAD** [33]. These are single-hop datasets whose answers typically exist in a single evidence paragraph, and they primarily test whether an adaptive method can correctly *skip* retrieval rather than orchestrate it. This thesis uses PopQA as the representative single-hop test set because it stresses factual recall and over-decomposition on entity-centric questions.

Chapter 3 Methodology

This chapter presents **SEEK** (**S**elf-certainty **E**valuated **E**vidence-gated **K**nowledge-retrieval), a lightweight adaptive RAG framework that treats question decomposition as a routing problem. SEEK runs both a direct path and a decomposition path, preserves original-question evidence inside decomposed steps, and chooses between the candidate answers using a training-free verifier. The design follows the view that adaptive RAG is more than a retrieve-or-skip problem: a question may need to be answered directly, decomposed into sharper retrieval queries, or repaired with original-question context after decomposition.

3.1 Overview

Given a question q and an open-domain retrieval corpus $\mathcal{P}$, SEEK produces a final answer $\hat{a}$ through two candidate-generation paths followed by a verifier. Figure 3.1 shows the overall pipeline:

1. **Direct path.** The answerer directly answers the original question q with BM25 evidence retrieved from $\mathcal{P}$, producing a direct candidate a^{dir} and a token-level confidence score σ^{dir} over the generated answer span.

2. **Decomposition path.** The decomposer $\mathcal{D}_\theta$ maps q to ordered sub-questions $\{s_1, \ldots, s_n\}$. Each sub-question is answered sequentially, with placeholder substitution, confidence probing, conditional retrieval, original-question evidence union, and a final synthesis step that produces a^{dec}.

3. **Training-free verifier.** A deterministic verifier compares a^{dir} and a^{dec} using query-level signals: direct-answer confidence, decomposition stability, post-synthesis confidence, and answer agreement. It returns either a^{dir} or a^{dec} as $\hat{a}$.

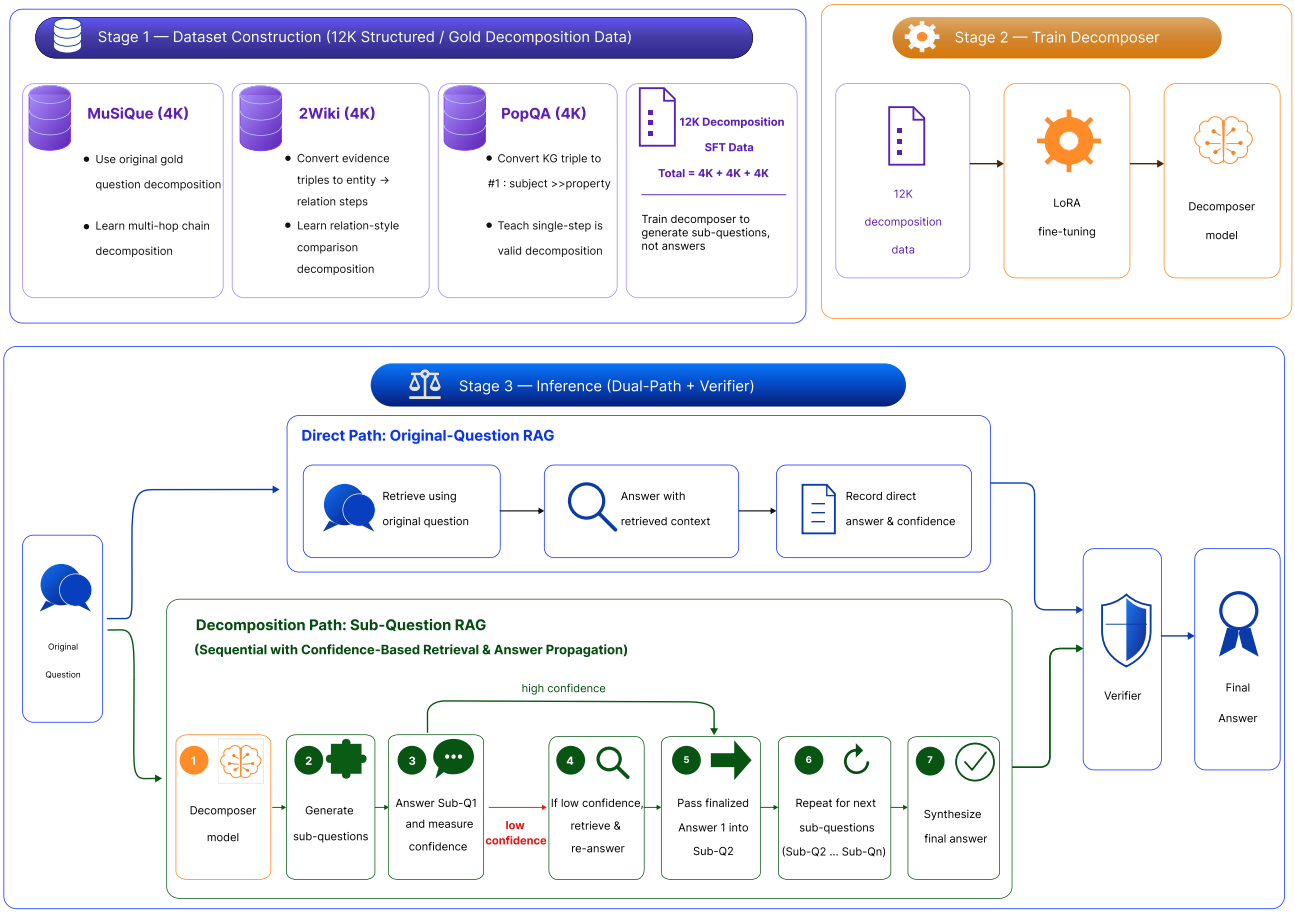


Fig. 3.1: Overview of SEEK. Stage 1 constructs 12K gold decomposition examples from MuSiQue, 2WikiMultihopQA, and PopQA; Stage 2 trains the LoRA decomposer on this corpus; Stage 3 runs dual-path inference, where the direct path answers the original question with retrieved evidence and the decomposition path answers sub-questions sequentially with confidence-gated retrieval and answer propagation, and a training-free verifier selects the final answer.

3.2 Problem Formulation

We study open-domain question answering: given a natural-language question q and a passage corpus $\mathcal{P}$, the system outputs a short answer span $\hat{a}$, which is evaluated against gold answers by exact match and F1. The questions of interest are often multi-hop, so a single retrieval with the original question may not surface all the required evidence. Let $\mathcal{M}_\theta$ be a pretrained language model, $r(q, \mathcal{P}, k)$ a BM25 retriever over an open-domain corpus $\mathcal{P}$, and $\mathcal{D}_\theta(q)$ a decomposer that emits sub-questions. A conventional retrieve-then-read system produces one answer from one path. SEEK instead constructs two candidates. The direct candidate answers the original question with retrieved evidence:

$$a^{\text{dir}} = \mathcal{M}_\theta(q, r(q, \mathcal{P}, k)).$$

The decomposition candidate first maps the question to an ordered list of sub-questions,

$$\{s_1, \ldots, s_n\} = \mathcal{D}_\theta(q),$$

then answers each sub-question s_i in order to obtain a_i, and finally synthesises the pairs into an answer span:

$$a^{\text{dec}} = \text{Synth}\left(q, \{(s_i, a_i)\}_{i=1}^{n}\right).$$

Here Synth is not a string-level merge but a final generation pass of the base model $\mathcal{M}_\theta$: it reads q together with all pairs (s_i, a_i) and generates the final answer span under the prompt rules described in Section 3.5.

The final decision is a verifier v:

$$\hat{a} = v\left(q, a^{\text{dir}}, a^{\text{dec}}, z^{\text{dir}}, z^{\text{dec}}\right),$$

where z^{dir} and z^{dec} are inference-time diagnostic signals such as confidence scores, unresolved placeholders, and candidate agreement. The verifier is a transparent rule set designed to approximate the oracle choice between direct and decomposed reasoning.

3.3 Sub-Question Decomposer

The decomposer emits one sub-question per line, prefixed with `#`$\langle i \rangle$`:`. Each sub-question is expressed as a compact knowledge-graph fragment:

$$\#i: \ \langle entity \rangle \ \texttt{>>} \ \langle relation \rangle$$

When a sub-question depends on an earlier answer, the placeholder `#`j refers to step j. For example:

```
#1: X >> director
#2: #1 >> place of birth
```

At inference, placeholders are resolved sequentially: once a_1 is produced, `#1` in s_2 is rewritten as a_1 before retrieval and answering. The KG-triple notation is compact, consistent across datasets, and aligned with Wikidata-style supervision in 2WikiMultihopQA and PopQA.

The decomposer is prompted with a fixed instruction:

> *Decompose the following question into an ordered list of atomic sub-questions. Each sub-question must be answerable in a single retrieval and reasoning step. When a sub-question depends on the answer of an earlier one, refer to that step with a `#N` placeholder. Output one sub-question per line, prefixed with `#<index>:`*

Table 3.1 shows representative decomposition patterns for each dataset.

Training. Only the decomposer requires training; the answerer, synthesizer, and verifier are training-free. The decomposer is a LoRA [34] adapter on Llama-3-8B-Instruct, trained on 12,000 examples drawn equally from MuSiQue gold sub-questions, 2WikiMultihopQA evidence-triple conversions, and PopQA one-step triples. The PopQA component teaches the decomposer that a one-step plan is valid. This improves single-hop robustness, but Chapter 4 shows that it also creates negative transfer on some multi-hop datasets; the verifier is introduced precisely because no single hard decomposition policy is optimal

Table 3.1: Representative decomposition patterns produced by the SEEK decomposer.

Dataset	Question type	Decomposition
PopQA	1-hop fact lookup	`#1: George Rankin >> occupation`
HotpotQA	2-hop bridge	`#1: Film X >> director` `#2: #1 >> nationality`
2WikiMultihopQA	comparison	`#1: A >> date of birth` `#2: B >> date of birth`

MuSiQue	compositional chain	`#1: X >> capital`
		`#2: #1 >> mayor`
		`#3: #2 >> party`

across all query types. At inference, the adapter is enabled for decomposition and sub-question answering, and disabled for final synthesis, where the base instruction model produces shorter and more stable spans. Adapter and optimisation hyperparameters are listed in Chapter 4.

3.4 Token-Level Confidence Probe

Both the direct path and each sub-question answer use an answer-only prompt:

> *Answer the question with the answer phrase only. No explanation, no preamble, no full sentence — just the answer.*

During greedy decoding, SEEK records the softmax distribution $p_t = \mathrm{softmax}(\mathrm{logits}_t)$ for each generated token t. Classical token-level confidence signals computed from these distributions include:

$$\sigma^{\text{max-ent}} = \max_t H(p_t), \tag{3.1}$$

$$\sigma^{\text{mean-ent}} = \frac{1}{T}\sum_t H(p_t), \tag{3.2}$$

$$\sigma^{\text{min-lp}} = \min_t \log p_t(y_t), \tag{3.3}$$

where $H(p) = -\sum_v p(v) \log p(v)$ and EOS tokens are excluded.

Tail-leakage surprisal. SEEK uses a different statistic, which we call *tail-leakage surprisal*, in both roles where confidence is read: the per-step retrieval gate and the query-level signals of the verifier. Let $V_5(p_t)$ be the set of the five most probable tokens under p_t, and let

$$m_t = \sum_{v \in V_5(p_t)} p_t(v)$$

be the probability mass the model keeps inside its own top-five candidates at position t. The per-token tail leakage is $\mathrm{TL}_t = -\log m_t$, the self-information of the event that decoding stays inside the candidate set at step t. The answer-level signal sums this leakage over the generated span:

$$\sigma^{\mathrm{tl}} = \sum_t \mathrm{TL}_t = -\log \prod_t m_t,$$

again with EOS tokens excluded. A value near zero means the whole answer was generated with essentially no probability mass escaping the top-five candidates, while larger values mean the model repeatedly placed belief on tokens outside the candidates it was choosing among. The per-token term also equals the KL divergence between the renormalised top-five truncation of p_t and p_t itself, so it measures how much of the model's belief lies outside the visible top of the distribution.

Why leakage instead of entropy. By the chain rule, token entropy splits into three parts:

$$H(p_t) = H_{\mathrm{B}}(m_t) + m_t\, H(\hat{p}_t^{\mathrm{in}}) + (1 - m_t)\, H(\hat{p}_t^{\mathrm{out}}),$$

where H_{B} is the binary entropy of the escape event and $\hat{p}_t^{\mathrm{in}}$, $\hat{p}_t^{\mathrm{out}}$ are the renormalised distributions inside and outside the top-five set. Only the first term reflects whether probability mass escapes the candidate set. The second term is often harmless for correct answers, because competition inside the top five frequently comes from aliases, casing, or formatting variants of the same answer, and the third term describes the composition of the tail, which carries little information about correctness. Tail-leakage surprisal keeps

the escape component and drops the two diluting terms. Summation matters as well: a wrong answer often leaks a small amount of mass at several positions rather than a large amount at one, and a maximum-based aggregate cannot accumulate this pattern. Chapter 4 compares tail-leakage surprisal against entropy and other confidence signals.

3.5 Decomposition-Path Inference

Within the decomposition path, retrieval remains sub-question-level. For a resolved sub-question s_i, the answerer first produces a tentative parametric answer $\tilde{a}_i$ and its tail-leakage surprisal σ_i. The retriever is invoked when:

$$\pi(s_i) = \mathbb{1}[\sigma_i \geq \tau],$$

with $\tau = 0.0085$ in the final system, chosen so that the most confident tenth of development sub-question steps fall below it and skip retrieval. If $\pi(s_i) = 1$, BM25 retrieves top-k passages and the answerer re-answers s_i with context; otherwise $\tilde{a}_i$ is kept. This local gate reduces unnecessary retrieval inside the decomposition path, but it is no longer the final routing mechanism for the whole question. The retrieval corpus, the per-dataset top-k, and the path-aware title policy are specified in Chapter 4.

Original-question evidence union. A decomposed sub-question is often shorter and more entity-centric than the original question, which improves retrieval precision but can remove global clues such as comparison targets or disambiguating conditions. To preserve this context, the final system augments each retrieved sub-question context with evidence retrieved from the original question:

$$C_i = r(s_i, \mathcal{P}, k) \cup r(q, \mathcal{P}, k).$$

The union is deduplicated before answering. This operation does not change the decomposer or train a new retriever; it simply keeps the original-question evidence available when answering each decomposed step. Although these original-question passages are also the evidence source for the direct candidate, the two answer paths remain distinct:

the direct path answers the original question directly, whereas the decomposition path still performs sub-question retrieval, step answering, and final synthesis. Thus the union shares evidence, not the direct answer or the final routing decision. Chapter 4 evaluates this operation as a decomposition-path ablation, especially on HotpotQA and 2WikiMultihopQA.

Synthesis. The decomposed answer a^{dec} is produced by a synthesis pass over the original question and all (s_i, a_i) pairs. The prompt encodes three answer-shaping rules derived from error analysis:

1. **Comparison questions** return the matching entity, not the compared value.
2. **Yes-or-no questions** return "Yes" or "No" based on agreement between sub-answers.
3. **Terse-span rule** strips unnecessary modifiers and keeps answer spans short for EM compatibility.

The implementation also includes a value-to-entity guard for comparison failures: if synthesis emits a value such as a year but the question asks which entity satisfies a comparison, the system maps the value back to the owning entity when this mapping is unambiguous.

Worked example. A 2WikiMultihopQA question from the final traces illustrates the full path. For *"Who is the mother of the director of film Polish-Russian War (Film)?"*, the decomposer emits `#1: Polish-Russian War >> director` and `#2: #1 >> mother`. The tentative answer for `#1`, produced without retrieval, is a hallucinated name, and its confidence signal indicates high uncertainty, so the gate retrieves and the step answers Xawery Żuławski. The placeholder in `#2` is then resolved, giving `Xawery Żuławski >> mother`; the tentative answer is again wrong, the gate retrieves again, and the step returns Małgorzata Braunek. Synthesis reads the original question together with the two resolved pairs and generates Małgorzata Braunek as a^{dec}. The direct path answers Dorota Masłowska, the author of the novel the film is based on, so the two candidates disagree and the verifier described in the next section selects the decomposed answer. In chain-style questions like this one the final answer often coincides with the last sub-answer; synthesis

carries real work mainly on comparison and yes-or-no questions, where it must select the winning entity or judge agreement between sub-answers.

3.6 Training-Free Verifier

The verifier chooses between the direct candidate and the decomposed candidate. It is intentionally conservative: decomposition remains the default because it improves multi-hop retrieval, and the verifier switches to the direct answer only when the direct path is confident and the decomposition path shows signs of instability. The final implementation additionally applies a conservative *type-aware fallback*. The fallback switches to the direct answer when the decomposed answer does not fit the question, for example a date for a question that asks for a person, or an answer that merely repeats an intermediate step answer; it does not rerun generation and does not train a router.

Signals. The verifier uses four query-level signals:

- **Direct-answer surprisal** σ^{dir}: low tail-leakage surprisal suggests the direct answer may be reliable.
- **Decomposition post-surprisal** σ^{dec}: high tail-leakage surprisal after decomposition and synthesis suggests the decomposed answer is unstable.
- **Structural validity**: unresolved placeholders, empty sub-answers, or malformed sub-questions indicate decomposition failure.
- **Candidate agreement**: normalised string agreement between a^{dir} and a^{dec} is treated as mutual support.

Rule. The verifier accepts agreement immediately; otherwise it chooses direct when the direct answer is confident and decomposition is unstable, and chooses decomposition in the remaining cases. Both verifier signals are tail-leakage surprisal, the same statistic used by the retrieval gate. The deployed thresholds are `direct_tl_tau=0.0005` and `decomp_post_tl_tau=0.0158`: the direct-side threshold admits roughly the most confident 12% of direct answers, and the decomposition-side threshold marks roughly the least

confident 57% of decomposed answers as unstable. The type-aware fallback is a deterministic refinement: it only changes the final route when answer type, direct surprisal, decomposition post-surprisal, and entity-fragment signals consistently indicate that the direct answer should be preferred. No dataset name, gold answer, trained classifier, or evidence-label supervision is used.

3.7 Implementation Summary

Algorithm 1 summarises the final inference loop.

Algorithm 1 SEEK v2 inference for one question q

Require: question q, corpus $\mathcal{P}$, surprisal threshold τ, top-k

$(a^{\text{dir}}, \sigma^{\text{dir}}) \leftarrow \textsc{DirectAnswer}(q, r(q, \mathcal{P}, k))$
$\{s_1, \dots, s_n\} \leftarrow \mathcal{D}_\theta(q)$
$A \leftarrow \emptyset$
for $i = 1$ to n **do**
    $s_i \leftarrow \textsc{Resolve}(s_i, A)$
    $(\tilde{a}_i, \sigma_i) \leftarrow \mathcal{A}_\theta(s_i, \emptyset)$
    if $\sigma_i \geq \tau$ **then**
        $c_i \leftarrow r(s_i, \mathcal{P}, k) \cup r(q, \mathcal{P}, k)$
        $a_i \leftarrow \mathcal{A}_\theta(s_i, c_i)$
    else
        $a_i \leftarrow \tilde{a}_i$
    end if
    $A[i] \leftarrow a_i$
end for
$(a^{\text{dec}}, \sigma^{\text{dec}}, z^{\text{dec}}) \leftarrow \textsc{Synthesize}(q, \{(s_i, a_i)\}_{i=1}^{n})$
$\hat{a} \leftarrow \textsc{Verifier}(a^{\text{dir}}, a^{\text{dec}}, \sigma^{\text{dir}}, \sigma^{\text{dec}}, z^{\text{dec}})$
return $\hat{a}$

The next chapter evaluates this final dual-path system against DeepRAG under the honest wiki-21M retrieval protocol.

Chapter 4 Experiments

This chapter evaluates the final SEEK system under the same honest wiki-21M retrieval protocol used by DeepRAG and compares it with representative reasoning and adaptive-retrieval baselines. After the main results, the chapter examines why the system works: whether decomposition actually improves retrieval, how much of the remaining MuSiQue gap comes from wrong intermediate answers rather than from decomposition wording, and whether retrieval or the confidence gate could be removed without losing accuracy. The later sections then look at the two answer paths in more detail. They measure how often each path is selected and where each path fails, compare SEEK's errors with DeepRAG's on the same questions, and ask how far tail-leakage surprisal can be trusted as a confidence signal.

4.1 Experimental Setup

4.1.1 Datasets

We evaluate on HotpotQA, 2WikiMultihopQA, MuSiQue, and PopQA. HotpotQA, 2WikiMultihopQA, and MuSiQue test multi-hop reasoning, while PopQA tests single-hop factual recall and exposes over-decomposition on simple entity-centric questions. To align with DeepRAG [14], all headline results use the first 1,000 questions of each development set without shuffling.

4.1.2 Retrieval Corpus

The main comparison uses the honest open-domain corpus: the shared Elasticsearch `wiki` index over `psgs_w100`, containing approximately 21M Wikipedia passages. We use BM25 with top-$k = 8$ for HotpotQA, 2WikiMultihopQA, and MuSiQue, and top-$k = 3$ for PopQA. SEEK applies a path-aware title policy: the direct path retrieves with the original question over paragraph text only, while the decomposition path retrieves short sub-question queries over title and paragraph text. This policy is dataset-agnostic because

it depends only on query type.

The final system further uses *original-question evidence union*. For each decomposed sub-question, the evidence context is the union of passages retrieved by the sub-question and passages retrieved by the original question. This preserves the global context that can be lost when a long question is decomposed into short sub-questions.

4.1.3 Metrics and Baselines

We report SQuAD-style exact match (EM) and token-level F1 after standard normalisation. When a dataset provides aliases, the maximum score over aliases is used. The main table includes representative Llama-3-8B baselines reported by DeepRAG [14]: CoT and CoT-Retrieve as reasoning baselines, FLARE [9] and DRAGIN [10] as training-free adaptive retrieval baselines, and DeepRAG as the strongest learned adaptive baseline. Internal comparisons include the path-aware baseline, the decomposition-path original-question evidence union ablation, and the final SEEK system with type-aware verifier replay.

4.1.4 Implementation Details

All SEEK results use Llama-3-8B-Instruct with the 12K three-source LoRA decomposer trained on MuSiQue, 2WikiMultihopQA, and PopQA. The adapter uses rank 16, alpha 32, and dropout 0.05 on all linear modules, trained in bf16 with a 1024-token context for 3 epochs using LLaMA-Factory [35], with effective batch size 16 and learning rate 1×10^{-4} under cosine decay with 10% warmup; training takes about 38 minutes on one A6000 GPU. At inference, the adapter is enabled for decomposition and sub-question answering and disabled for synthesis, and the decomposition path uses the tail-leakage surprisal gate at $\tau = 0.0085$. The final verifier uses the rule set of Chapter 3 with tail-leakage surprisal as its confidence signal, plus a conservative type-aware fallback. This fallback changes only the final choice between the direct and decomposed candidates; it does not rerun generation and does not train an additional model.

Several ablations and diagnostic analyses in this chapter were carried out on development traces produced while the system was being built, some of them before the gate and verifier signals changed from maximum token entropy to tail-leakage surprisal. Their

absolute numbers can therefore differ slightly from the final configuration, but each table is internally consistent and the comparisons within it are unaffected.

4.2 Main Results

Table 4.1 reports the final honest wiki-21M results. To keep the comparison aligned with the SEEK evaluation protocol, the table shows the four overlapping datasets used in this thesis and omits CAG and WebQuestions, which were reported by DeepRAG but not evaluated by SEEK. The final SEEK system combines path-aware title retrieval, original-question evidence union, surprisal-gated sub-question retrieval, and the type-aware training-free verifier.

Table 4.1: Main results on the first 1,000 questions of each dev set using the honest wiki-21M pool. Each cell reports EM/F1.

Type	Method	HotpotQA	2Wiki	MuSiQue	PopQA	Average
Reasoning	CoT	27.20/37.75	28.20/34.85	13.70/22.97	21.20/25.33	22.58/30.23
Reasoning	CoT-Retrieve	34.90/46.85	35.80/43.41	19.10/28.18	32.80/45.87	30.65/41.08
Adaptive	FLARE [9]	23.80/32.88	30.30/37.45	14.50/23.57	28.80/40.61	24.35/33.63
Adaptive	DRAGIN [10]	27.60/38.05	29.10/35.68	11.80/19.97	22.60/28.53	22.78/30.56
Learned adaptive	DeepRAG [14]	**40.70/51.54**	48.10/53.25	22.50/30.40	42.50/47.80	38.45/45.75
Ours	**SEEK final**	37.50/48.72	**48.70/55.63**	**26.60/34.74**	**53.70/55.53**	**41.62/48.66**

Note. External baseline numbers are the Llama-3-8B results reported by DeepRAG [14]. The average columns here are recomputed over the four datasets evaluated by SEEK.

The final SEEK system reaches 41.62 EM and 48.66 F1 on the four-dataset average, compared with DeepRAG's 38.45 EM and 45.75 F1, a gain of +3.17 EM and +2.91 F1 over the strongest external baseline. The margin over the reasoning baselines and the training-free adaptive baselines is considerably wider.

At the per-dataset level, SEEK wins on three of the four datasets. On 2Wiki it improves from DeepRAG's 48.10 EM to 48.70 EM, and on MuSiQue from 22.50 to 26.60 EM. The largest gain is on PopQA, where SEEK reaches 53.70 EM against DeepRAG's 42.50, which suggests the direct-vs-decompose verifier is especially useful when over-decomposition is risky.

HotpotQA is the exception. SEEK improves it from the path-aware baseline of 31.80 EM to 37.50 EM but still trails DeepRAG's 40.70 EM. HotpotQA is therefore the main remaining limitation and motivates the analysis in Sections 4.5.2–4.5.5.

4.3 Retrieval Analysis

This section asks whether decomposition improves retrieval itself, and how much of the improvement depends on the trained decomposer rather than on decomposition as such.

4.3.1 Answer-Containing Recall

Before evaluating final answers, we first ask whether decomposition actually improves retrieval quality. Table 4.2 reports answer-containing hit@5: a query is counted as successful if at least one of the top five retrieved passages contains the gold answer or one of its aliases. Yes-or-no questions are excluded because the strings "yes" and "no" are not meaningful evidence strings.

Table 4.2: Answer-containing hit@5 under the final open-domain wiki retrieval condition. Original-query retrieval uses paragraph text only; decomposed sub-question retrieval uses title and paragraph text.

Dataset	*N*	Original	Decomposed	Union	Decomp − Original
2Wiki	917	31.73	62.38	**65.32**	+30.64
HotpotQA	941	46.55	56.54	**65.25**	+9.99
MuSiQue	1000	16.90	37.40	**41.60**	+20.50
PopQA	1000	68.10	68.70	**73.30**	+0.60

Decomposition substantially improves answer-containing retrieval on the multi-hop datasets: +30.64 points on 2Wiki, +20.50 on MuSiQue, and +9.99 on HotpotQA. PopQA is nearly unchanged because it is already single-hop. The union column is strongest on all four datasets, showing that decomposition should not discard the original question's global context.

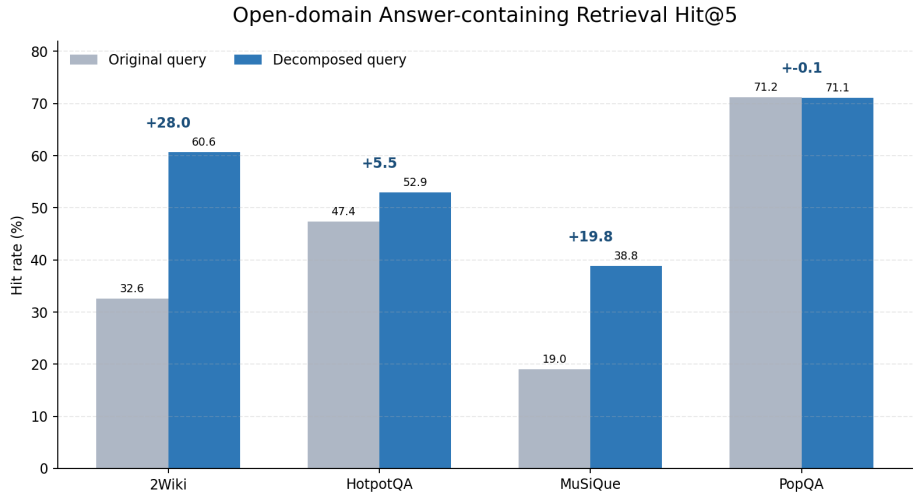


Fig. 4.1: Answer-containing retrieval recall under the open-domain wiki-21M condition. Decomposition improves multi-hop retrieval, while original-question evidence union further recovers context.

4.3.2 Gold and Base Decomposition on MuSiQue

To separate the value of decomposition from the value of the trained decomposer, we run a focused MuSiQue retrieval-recall analysis using the first 1,000 development questions. This slice contains only 2-hop questions, which makes the official MuSiQue gold decomposition directly comparable with SEEK's predicted two-step chain. Table 4.3 reports answer-containing hit@5 under the same wiki-21M and path-aware retrieval condition as Table 4.2.

The gap between original retrieval and SEEK retrieval is large: 19.0 to 38.8 hit@5. The oracle gold chain reaches 55.8, showing that decomposition has further retrieval headroom. However, the important diagnostic is that *Gold raw* collapses back to the original-query level. A 2-hop decomposition is useful only when the first-hop bridge entity is correctly resolved and substituted into the second query. SEEK's first-hop intermediate-answer EM is 41.9%. When the first hop is answered correctly, SEEK reaches 56.4 hit@5, essentially matching the gold chain on the same subset at 58.5. When the first hop is wrong, SEEK falls to 25.9 while gold stays at 53.8, so these questions are not intrinsically harder and their evidence remains reachable with the right bridge entity. The SEEK–gold

Table 4.3: MuSiQue answer-containing hit@5 for original, SEEK, gold, and untuned-base decompositions.

Arm	Interpretation	Hit@5
Original question	No decomposition	19.0
SEEK	SEEK decomposition with predicted intermediate answer	38.8
Gold	Official MuSiQue decomposition with gold intermediate answer	**55.8**
Gold raw	Official decomposition without replacing `#1`	19.8
SEEK + gold bridge	SEEK sub-question text with gold intermediate answer	54.1
Untuned base + gold bridge	Llama-3-8B-Instruct without LoRA, using gold intermediate answer	52.9

gap is therefore almost entirely bridge-answer drift rather than poor decomposition wording.

The untuned base model also clarifies why the decomposer is trained. With the same prompt and no LoRA adapter, the base model produces a valid two-step decomposition for only 149 of 1,000 MuSiQue examples and has a 15.3% format-parsing failure rate. Its query text can retrieve well under oracle bridge substitution, but it is not structurally reliable enough to deploy. LoRA training therefore buys step-count control, valid placeholders, and compact BM25-friendly query style, not merely better wording.

4.4 Ablation Studies

This section removes or varies one design component at a time: retrieval after decomposition, decomposition itself, the gate threshold, the title-matching policy, and the original-question evidence union.

4.4.1 Retrieval Is Necessary After Decomposition

Decomposition alone does not produce factual knowledge. Table 4.4 compares three decomposition-path settings: never retrieve, gated retrieval, and always retrieve. This ab-

lation predates the switch to tail-leakage surprisal and uses the development-stage entropy gate at $\tau = 0.3$; the replay in Section 4.4.3 shows the same pattern for the final gate.

Table 4.4: Ablation of retrieval after decomposition under honest wiki-21M. EM is reported for each answering strategy; retrieval rate is measured inside the decomposition path.

Dataset	Never retrieve	Gate $\tau = 0.3$	Always retrieve	Gain	Retrieval rate
HotpotQA	15.7	25.0	25.1	+9.3	92.7%
2Wiki	29.4	47.2	47.1	+17.8	95.2%
MuSiQue	9.6	24.7	25.0	+15.1	90.8%
PopQA	24.2	51.5	51.4	+27.3	99.3%

Retrieval after decomposition is necessary: the gate improves over never retrieve by +9.3 to +27.3 EM. However, the gate and always-retrieve settings are almost identical in accuracy. This positions confidence gating as an efficiency mechanism rather than the main source of accuracy.

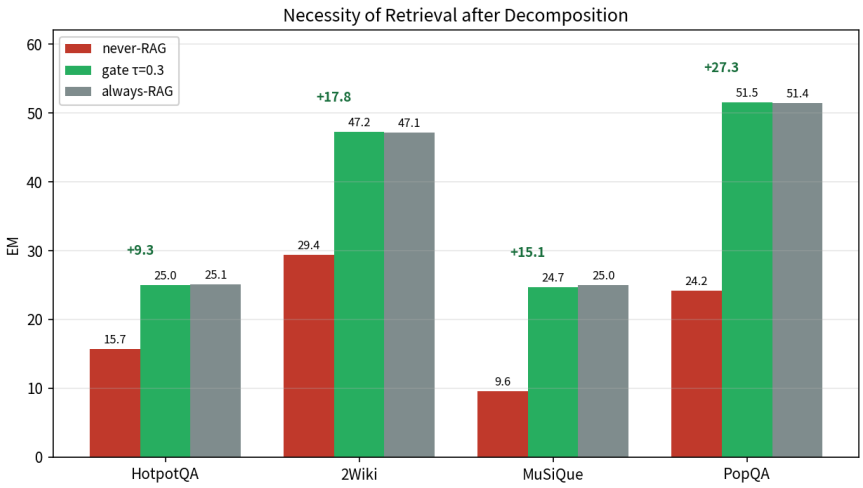


Fig. 4.2: Three-way decomposition-path ablation: never retrieve, gated retrieval, and always retrieve. The gate nearly matches always-retrieve accuracy while reducing some retrieval calls.

4.4.2 Direct-Only Confidence-Gated Baseline

The previous ablation shows that retrieval is necessary after decomposition. A separate question is whether decomposition itself is necessary, or whether the same confidence gate could simply control retrieval on the original question. Table 4.5 compares a direct-only confidence-gated RAG baseline against the decomposition-path gate. The direct baseline first answers the original question without context, measures the uncertainty of the draft answer, and retrieves the original question only when the uncertainty exceeds a threshold. Both arms of this development ablation use the same entropy gate, so the comparison isolates decomposition rather than the choice of signal. The table gives the matched threshold $\tau = 0.3$ and the best threshold selected retrospectively for the baseline.

Table 4.5: Direct-only confidence-gated RAG compared with decompose+gate. EM is reported under honest wiki-21M.

Dataset	Direct never	Direct gate $\tau = 0.3$	Direct best τ	Decompose+gate	Decompose − direct best
HotpotQA	18.2	**30.7**	30.7	25.0	-5.7
2Wiki	12.6	18.5	18.6	**47.2**	+28.6
MuSiQue	5.9	10.4	10.8	**24.7**	+13.9
PopQA	22.4	37.1	39.2	**51.5**	+12.3

Confidence-gated retrieval without decomposition plateaus at the single-query retrieval ceiling. Even when allowed to choose the best threshold after the fact, it remains far below decompose+gate on 2Wiki, MuSiQue, and PopQA. HotpotQA is the exception because its original questions often already contain enough lexical signal for BM25 to retrieve useful evidence. The final SEEK system handles this case by keeping the direct path and routing late; the final system reaches 37.5 EM on HotpotQA, above the direct-gate baseline.

4.4.3 Gate Threshold Sensitivity

Table 4.6 evaluates the final dual-path system under different tail-leakage gate thresholds, replayed offline from the stored per-step signals. Higher thresholds reduce retrieval,

but the average score declines.

Table 4.6: Final SEEK performance across tail-leakage gate thresholds. Each score cell reports average EM; F1, and retrieval rate is the average decompose-path retrieval rate.

τ	Average EM; F1	Retrieval rate
0.0023	40.58; 47.62	95.5%
0.0085	**40.60**; **47.69**	90.8%
0.0222	40.25; 47.33	85.9%
0.0442	39.83; 46.84	81.1%
0.1165	38.60; 45.50	71.2%

The deployed threshold $\tau = 0.0085$ skips the most confident tenth of decomposed steps. It removes about nine points of retrieval rate at no accuracy cost and is also the best-scoring point of the sweep. Tightening the gate further trades accuracy for retrieval savings: $\tau = 0.0442$ saves another ten points of retrieval rate but costs 0.77 average EM. The curve supports a conservative interpretation: under honest wiki-21M, most decomposed steps still require retrieval, and tightening the gate eventually hurts multi-hop datasets such as 2Wiki and MuSiQue.

4.4.4 Path-Aware Title Retrieval

Original questions and decomposed sub-questions are different query types. A short query such as `Houston Baptist University founded` benefits from title matching because the title often identifies the relevant Wikipedia article. A long original question, however, can contain multiple entities and conditions; title matching can overemphasise a stray entity and retrieve the wrong article. Table 4.7 compares title-field policies under the same verifier setting.

Path-aware title retrieval is the most stable option. It keeps the title benefit for short sub-questions while avoiding the HotpotQA drop caused by enabling title matching on long original questions.

Table 4.7: Title-field matching ablation on honest wiki-21M. Each cell reports EM; F1.

Dataset	Off	Global	Path-aware
PopQA	51.2; 53.5	51.5; 53.9	**51.8**; **54.2**
MuSiQue	22.5; 30.5	24.5; 33.1	**24.6**; **33.5**
2Wiki	45.2; 51.3	45.9; 52.5	**46.9**; **53.5**
HotpotQA	31.9; 41.2	30.5; 39.9	**31.8**; **41.2**

4.4.5 Original-Question Evidence Union

The preceding retrieval results suggest that decomposition improves precision but may lose global context. Original-question evidence union addresses this by adding the original-question top-k evidence to each decomposed step. Since this operation only changes the decomposition path, the clean ablation compares the decomposed answer candidate with and without union, rather than mixing in questions for which the final verifier selected the direct path. Table 4.8 reports this path-local effect.

Table 4.8: Decomposition-path effect of adding original-question evidence to each decomposed sub-question context. Each cell reports EM/F1.

Dataset	Decomp w/o union	Decomp + union	Change
HotpotQA	25.00 / 33.42	**30.50 / 39.49**	+5.50 / +6.07
2Wiki	47.20 / 53.63	**50.20 / 56.65**	+3.00 / +3.03
MuSiQue	24.70 / 33.10	**25.50 / 33.40**	+0.80 / +0.30
PopQA	51.50 / 53.68	**52.90 / 55.20**	+1.40 / +1.52
Average	37.10 / 43.46	**39.78 / 46.19**	+2.68 / +2.73

The path-local improvement is largest on HotpotQA and 2Wiki, precisely where short sub-questions can remove comparison targets or entity constraints from the original question. The full-system score also improves after union, but that number includes the verifier's direct-vs-decompose choices; it is therefore a final-system result, not the clean isolation of the union operation. In short, decomposition should produce sharper sub-queries

without forcing the lower path to forget the original question.

4.5 The Two Answer Paths

The final system keeps two answer paths, so the remaining analyses ask how the paths divide the work: how often each is selected, where each fails, how the failures compare with DeepRAG's on the same questions, and how much routing headroom remains.

4.5.1 Final Path Selection Rate

Original-question evidence union shares original-question passages with the decomposition path, but it does not collapse SEEK into a single path. The final verifier still selects either the direct answer candidate or the decomposed answer candidate. Table 4.9 reports the selected path rates in the final traces.

Table 4.9: Final verifier path selection rate after original-question evidence union and type-aware fallback.

Dataset	Direct path selected	Decomposition path selected
HotpotQA	274 / 1000 (27.4%)	726 / 1000 (72.6%)
2Wiki	65 / 1000 (6.5%)	935 / 1000 (93.5%)
MuSiQue	182 / 1000 (18.2%)	818 / 1000 (81.8%)
PopQA	37 / 1000 (3.7%)	963 / 1000 (96.3%)
Pooled	558 / 4000 (14.0%)	3442 / 4000 (86.1%)

These rates show that the system remains decomposition-dominant overall, while still preserving a direct path for cases where decomposition loses important context or produces an unstable answer. The shared original-question evidence is a context-preservation mechanism; the answer candidates and final routing decision remain path-specific.

4.5.2 Direct-vs-Decompose Correctness Distribution

Table 4.10 compares answer correctness across the direct and decomposition paths in the final traces. This analysis asks whether decomposition is always better, or whether

the two paths solve different questions.

Table 4.10: Answer-level correctness distribution across direct and decomposition paths in the final traces.

Dataset	Both correct	Direct only	Decomp only	Both wrong	Decomp only − Direct only
HotpotQA	192	111	113	584	+2
2Wiki	141	39	361	459	+322
MuSiQue	71	31	184	714	+153
PopQA	350	28	179	443	+151

The distributions show that decomposition is useful but not universally dominant. On 2Wiki, MuSiQue, and PopQA, the decompose-only column is much larger than the direct-only column, confirming the value of decomposition. HotpotQA is different: direct-only and decompose-only are almost tied. This explains why HotpotQA remains difficult and why the final system must preserve both paths. The large both-wrong counts also show that many errors cannot be fixed by routing alone; they require better retrieval, reranking, decomposition quality, or final synthesis.

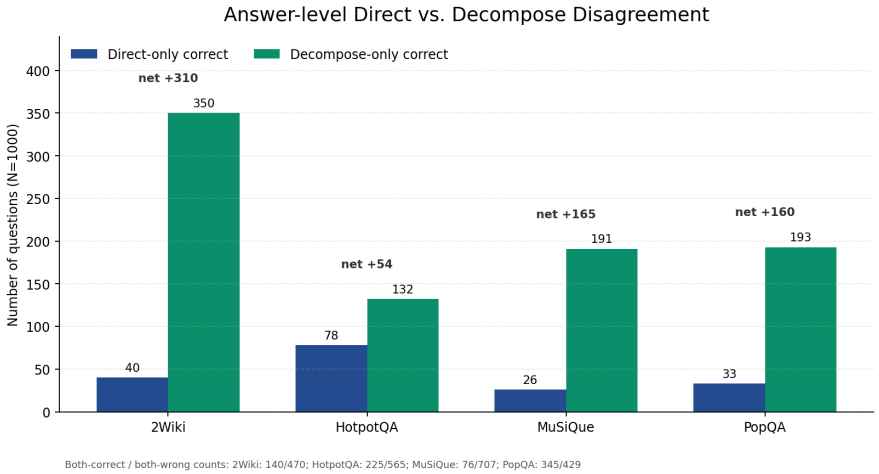


Fig. 4.3: Direct-vs-decompose correctness counts in the final traces. Decompose-only wins dominate on 2Wiki, MuSiQue, and PopQA, while HotpotQA is balanced.

4.5.3 Decomposition-Path Failure Taxonomy

To understand what remains after applying decomposition, evidence union, and verifier routing, we classify decomposition-path errors on HotpotQA, 2Wiki, and MuSiQue. Table 4.11 reports the primary error category among examples where the decomposed candidate is wrong.

Table 4.11: Failure taxonomy for decomposition-path errors. Values are shares within decomposed-candidate errors.

Category	HotpotQA	2Wiki	MuSiQue
Retrieval gap in both original and decomposed evidence	47%	64%	68%
Answer wrong despite retrieval coverage	28%	18%	12%
Synthesis failure	15%	17%	18%
Constraint loss from decomposition	10%	1%	2%

The dominant failure is not malformed decomposition; it is retrieval coverage. Neither the original query nor the decomposed evidence union reaches enough supporting information in 47% to 68% of decomposed-candidate errors. Constraint loss is more dataset-specific: it is visible on HotpotQA, where short KG-style sub-questions can drop the comparison target or disambiguating condition. This explains why original-question evidence union helps HotpotQA most in Table 4.8. Synthesis failure remains a stable 15% to 18% slice, motivating the answer-type and value-to-entity guards described in Chapter 3.

4.5.4 Case-Level Comparison with DeepRAG

The taxonomy above describes SEEK's own decomposition failures. A natural follow-up question is how these failures relate to DeepRAG's on the same inputs. We join the DeepRAG traces with the final SEEK traces by question text on the three multi-hop datasets and compare per-question EM. Table 4.12 reports the resulting counts.

The first observation is that the two systems are complementary rather than ordered. The DeepRAG-only and SEEK-only columns are of similar size on every dataset, so the

Table 4.12: Per-question correctness of DeepRAG and SEEK on the same first 1,000 dev questions. The SEEK-only column is analysed in Table 4.13.

Dataset	Both correct	DeepRAG only	SEEK only	Both wrong
HotpotQA	257	139	98	506
2Wiki	340	162	160	338
MuSiQue	154	83	92	669

comparison does not show one method dominating the other per question. Its value lies in showing that the two systems fail in different ways.

To characterise those differences, we inspect the DeepRAG traces inside the SEEK-win set and count recurring failure signals. Table 4.13 reports the counts; a single case can match several signals.

Table 4.13: Failure signals observed in DeepRAG traces on questions that SEEK answers correctly and DeepRAG does not. Counts are non-exclusive.

Signal	HotpotQA (n=98)	2Wiki (n=160)	MuSiQue (n=92)
Format near-miss, F1 ≥ 0.5	21	17	14
Degenerate final answer	24	29	27
Over-decomposition, ≥5 steps	5	39	5
Decomposition spiral	31	11	24
No retrieval in the whole chain	13	55	0

Three patterns account for most of the substantive losses. The first is over-decomposition on 2Wiki comparison questions, where 39 of the 160 SEEK wins involve DeepRAG chains of five or more steps. A representative case asks which film has the earlier-born director, *Buffet Froid* or *Lost in Africa*. DeepRAG produces six steps but retrieves only once: it resolves the first director correctly, then answers the remaining steps from parametric memory, hallucinates the second director, and finally appends two redundant comparison steps that restate the original question. The wrong parametric comparison in those terminal steps flips an otherwise recoverable chain. SEEK's decomposer emits exactly the four

factual lookups, retrieves for each, and leaves the comparison to the synthesis stage.

The second pattern is a decomposition spiral. DeepRAG generates each follow-up question conditioned on the chain answered so far, so a wrong intermediate answer can rewrite the rest of the plan. In one HotpotQA case, the model resolves the stage name Aladin to the wrong person and then spends five further steps trying to make that person fit the question, with later follow-up fields visibly containing fragments of the earlier erroneous reasoning. This contamination appears in roughly a quarter to a third of the SEEK-win cases on HotpotQA and MuSiQue. SEEK's decomposition is produced up front from the original question alone, so a wrong bridge answer degrades one retrieval query but cannot corrupt the structure of the remaining sub-questions.

The third pattern is structural mis-decomposition. Asked who sings *Home Alone Tonight* with the singer of another song, DeepRAG splits the question into two independent singer lookups, drops the duet relation, and returns the same artist for both. SEEK's placeholder binding keeps the relation explicit in the second sub-question and recovers the duet partner. On 2Wiki, an additional dataset-specific weakness appears: in 55 of the 160 SEEK wins, DeepRAG's retrieve-or-answer policy chose the parametric option at every step and hallucinated the whole chain. SEEK's tail-leakage gate keeps step-level retrieval near or above 90% on the multi-hop datasets (Table 4.4), so a fully parametric chain is rare.

These counts should be read with two caveats. About 15% to 21% of the SEEK-win set are format near-misses in which DeepRAG's answer is essentially correct but fails exact match, such as *Murray Hill, New Jersey* against the gold *Murray Hill*. The comparison is also symmetric: within its own win set, SEEK's decomposition is sometimes poor as well, and the question is rescued by the direct path or by original-question evidence union rather than by the decomposition itself. On HotpotQA, 29 of the 98 SEEK wins come from the direct candidate. The DeepRAG-only column contains failures of the mirrored kind, which is consistent with the dual-path design rather than evidence against it.

4.5.5 Oracle Direct-vs-Decompose Headroom

Table 4.14 reports an oracle that chooses between the direct and decomposed answer using the gold answer. This oracle is not deployable, but it quantifies the remaining

headroom in the routing decision.

Table 4.14: Oracle choice between direct and decompose in the final traces. Current router is the final SEEK verifier output.

Dataset	Direct EM	Decomp EM	Current router	Oracle	Oracle gap
HotpotQA	30.30	30.50	35.50	**41.60**	+6.10
2Wiki	18.00	50.20	50.00	**54.10**	+4.10
MuSiQue	10.20	25.50	24.70	**28.60**	+3.90
PopQA	37.80	52.90	51.90	**55.70**	+3.80

The oracle gap gives a precise interpretation of the remaining limitation. SEEK has not completely solved direct-vs-decompose routing; rather, it exposes the problem and recovers part of the available headroom with a training-free verifier. HotpotQA has the largest gap, +6.10 EM, because neither fixed path dominates.

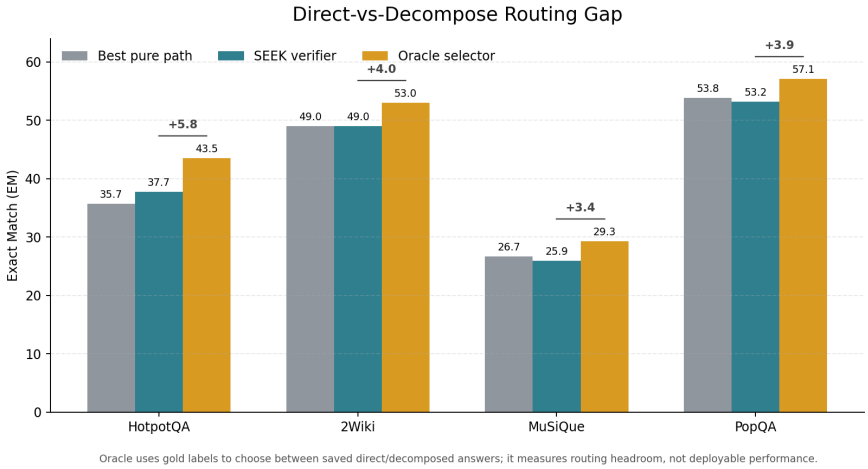


Fig. 4.4: Oracle direct-vs-decompose headroom in the final traces. The gap measures routing headroom only; both-wrong examples remain outside the verifier's reach.

4.6 Tail-Leakage Surprisal as a Confidence Signal

Tail-leakage surprisal is the signal the deployed gate relies on, so the final analysis asks how far it can be trusted. In the offline gate replay it is the strongest of the tested retrieval-gate signals: at the deployed threshold, which skips the most confident tenth of decomposed steps, it keeps the four-dataset average at 40.60 EM, while a maximum-entropy gate at the matched retrieval rate reaches 40.45. These results support tail-leakage surprisal as a sub-question-level retrieval gate.

The difference is clearer when the signals are scored as error detectors. We ran a battery of more than forty confidence signals on the direct path in the deployed setting, covering temperature-scaled and Rényi entropies, log-probability statistics, top-k mass and margin scores, self-consistency agreement, P(True)-style self-evaluation, and z-scored combinations, evaluated on the full 1,000-question traces of all four datasets with paired bootstrap over 1,000 resamples. Table 4.15 lists the strongest signals, scored by the AUROC of predicting a wrong direct answer. Tail-leakage surprisal has the best macro AUROC at 0.749 and beats the strongest alternative, minimum top-five mass, on every dataset, with the paired difference significant at $P < 0.001$. Temperature-2 entropy is stronger on 2Wiki and PopQA but collapses to 0.636 on MuSiQue, while maximum token entropy stays at 0.726.

Table 4.15: Strongest signals from the direct-path confidence battery on the full 1,000-question traces of all four datasets. Each cell is the AUROC of predicting a wrong direct answer.

Signal	HotpotQA	2Wiki	MuSiQue	PopQA	Macro
Tail-leakage surprisal	**0.757**	0.717	**0.726**	0.798	**0.749**
Minimum top-5 mass	0.752	0.706	0.723	0.787	0.742
Rényi-0.5 max entropy	0.747	0.716	0.681	0.802	0.737
Temperature-2 max entropy	0.734	**0.723**	0.636	**0.818**	0.728
Maximum token entropy	0.743	0.671	0.711	0.780	0.726
Summed log-probability	0.733	0.667	0.707	0.798	0.726

The battery also gives the decomposition argument of Chapter 3 an empirical check.

Scoring each chain-rule component of token entropy separately, the escape term alone reaches a macro AUROC of 0.742, ambiguity within the top-five candidates reaches 0.712, and the composition of the tail reaches 0.502, barely above chance. Full entropy mixes the three and lands at 0.726, while summing the escape surprisal over the answer reaches 0.749. Summation contributes the rest of the margin: on single-token answers the sum and the weakest-token maximum coincide, but on answers of seven or more tokens the sum reaches an AUROC of 0.805 while the weakest-token aggregate stays at 0.788, because wrong answers tend to leak small amounts of mass at several positions rather than a large amount at one.

At the same time, tail-leakage surprisal is not a correctness guarantee. Across the pooled traces, 491 wrong direct answers carry surprisal at or below the median of correct answers: 22% of the wrong answers on 2Wiki, 17% on HotpotQA, 16% on MuSiQue, and 7% on PopQA. More than half of these confidently wrong answers are clean parametric hallucinations produced with essentially zero leakage, and another 148 are alias or boundary near-misses whose answer is essentially right but fails exact match. No threshold can reach the first group, because nothing in the token distribution marks the answer as wrong.

Table 4.16 shows the same limitation from the selective-answering side: questions are ranked by surprisal, only the most confident fraction is kept, and accuracy is measured inside the kept set.

Table 4.16: Direct-path accuracy when only the most confident fraction of answers is kept, ranked by tail-leakage surprisal. The 100% row is the unfiltered direct-path accuracy.

Coverage	HotpotQA	2Wiki	MuSiQue	PopQA	Pooled
25%	57.6	34.0	22.4	79.2	52.8
50%	45.4	26.8	15.2	57.0	38.9
75%	38.0	22.1	12.5	46.3	30.2
100%	30.3	18.0	10.2	37.8	24.1

Accuracy rises steadily as coverage shrinks, so the ranking is real; if the most confident answers were lucky guesses, the top quarter would not be systematically more accurate. The absolute levels are more sobering. On MuSiQue the most confident quarter of

direct answers is correct only 22.4% of the time, and on 2Wiki 34.0%, so on the multi-hop datasets even the most confident direct answers are far from trustworthy. Only single-hop PopQA reaches a usable 79.2%. The mirror group deserves equal care: 164 correct answers, 17% of all correct ones, carry surprisal at or above the median of wrong answers, so high surprisal is an equally poor reason to discard an answer.

The verifier is the one place where the final system still reads entropy. Its rescue rule compares query-level direct-answer entropy and decomposition post-entropy against tuned thresholds, and Table 4.17 shows that this signal remains serviceable at that level. It compares question-level direct-path entropy between examples where both paths are correct and examples where both paths are wrong: the pooled AUROC is 0.782, the mean rises from 0.485 to 1.404, and the last column counts the both-wrong examples whose entropy still sits at or below the both-correct median.

Table 4.17: Direct-path entropy separation between both-correct and both-wrong examples. Lower entropy means higher confidence.

Dataset	Mean entropy both-correct	Mean entropy both-wrong	AUROC	Confidently wrong rate
HotpotQA	0.407	1.151	0.750	18.3%
2Wiki	0.796	1.495	0.688	28.8%
MuSiQue	0.742	1.689	0.752	16.5%
PopQA	0.351	1.185	0.819	9.5%
Pooled	0.485	1.404	0.782	–

The AUROC of 0.782 should be read as a population-level statistic rather than as the accuracy of a per-question detector. The medians are well separated, with pooled values of 0.27 for both-correct and 1.12 for both-wrong, but the tails overlap heavily: on every dataset, the tenth percentile of the both-wrong group lies below the median of the both-correct group. This also explains why moving the verifier onto the new signal changes little: replacing the development-stage entropy thresholds with tail-leakage thresholds shifts the four-dataset average by less than 0.2 EM, within the replay noise, so the final system adopts the tail-leakage version mainly to keep a single confidence statistic throughout. The ranking gain of tail-leakage surprisal lies in the uncertain tail of the answer distri-

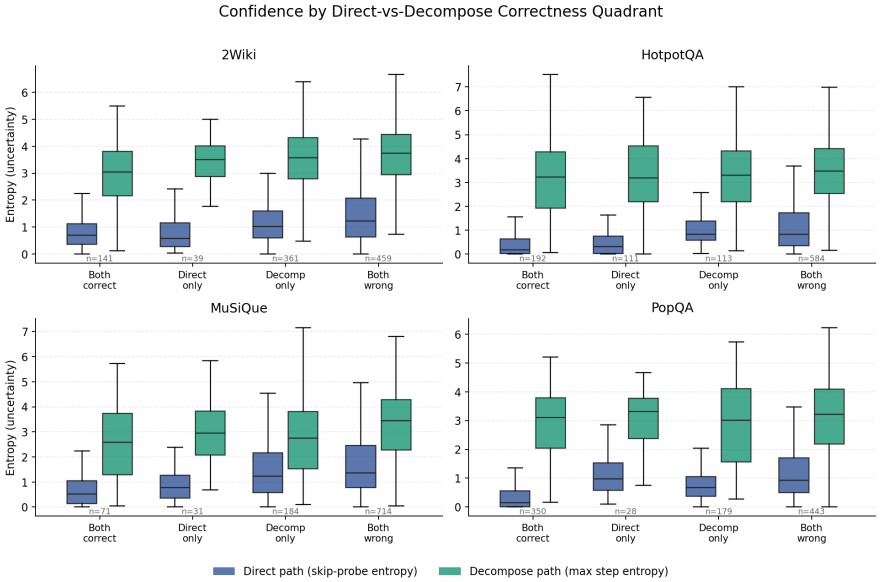


Fig. 4.5: Entropy by direct-vs-decompose correctness quadrant. Both-correct examples are more confident on average, but low entropy does not eliminate both-wrong cases.

bution, while the verifier's binary rules act on the confident head, where the two signals agree.

The decomposition-path signal is considerably weaker at the question level. Rerunning the battery on the decomposed answers gives tail-leakage surprisal a macro AUROC of only 0.631, essentially tied with the best alternatives, and on 2Wiki the strongest signal reaches only 0.587, close to random, because many 2Wiki parametric errors are produced with full confidence. The step-level signal remains informative as a retrieval gate, which is how SEEK uses it, but it says little about whether the final decomposed answer is right.

This is why SEEK deploys tail-leakage surprisal as a per-step retrieval gate and keeps confidence as only one signal inside a dual-path verifier, rather than treating any confidence score as a global router. Confidence helps estimate whether a path may be reliable; it does not reliably identify which path is correct when the two paths disagree.

4.7 Summary

Under the honest wiki-21M protocol, SEEK exceeds DeepRAG on the four-dataset average and wins on 2Wiki, MuSiQue, and PopQA, with HotpotQA remaining the only

dataset below DeepRAG. Decomposition is the main source of this result. It raises answer-containing recall strongly on the multi-hop datasets, and the gold MuSiQue decomposition shows that the recall ceiling extends to 55.8 hit@5, with most of the remaining gap explained by wrong intermediate answers rather than by decomposition wording. Retrieval stays necessary after decomposition, since removing it costs 9.3 to 27.3 EM, and the confidence gate cannot take decomposition's place: a direct-only confidence-gated baseline stays 12.3 to 28.6 EM below decompose-with-gate on 2Wiki, MuSiQue, and PopQA even with a retrospectively chosen threshold.

The analyses also mark the limits of each component. The confidence gate behaves as an efficiency mechanism rather than a correctness guarantee: gated retrieval nearly matches always-retrieve accuracy while saving some retrieval calls. Tail-leakage surprisal ranks wrong answers better than entropy and every other signal in a battery of more than forty, yet confidently wrong answers survive any single threshold. The direct and decomposition paths are complementary, with oracle routing worth a further 3.8 to 6.1 EM. The case-level comparison with DeepRAG shows the same complementarity across systems: DeepRAG's chain-conditioned decomposition suffers from over-decomposition, error spirals, and zero-retrieval hallucination that SEEK's up-front decomposition and near-always sub-question retrieval avoid, while DeepRAG wins a comparable number of questions of its own. Both-wrong examples remain the largest error mass on every dataset and call for better retrieval and synthesis rather than better routing alone.

Chapter 5 Discussion

This chapter interprets the final results from Chapter 4. Taken together, they argue against reducing adaptive RAG to a single retrieve-or-skip decision. In multi-hop QA, the system must also decide when to trust decomposition, when to preserve the original question, and when the remaining error is outside the reach of routing.

5.1 Cost–Accuracy Trade-off versus DeepRAG

DeepRAG [14] and SEEK both operate at a finer granularity than standard retrieve-then-read RAG, but the final experiments require a careful cost claim. SEEK should not be described as uniformly cheaper at inference time. In the retrieval-count audit over the honest wiki-21M traces, SEEK retrieves more often than DeepRAG on HotpotQA and 2Wiki, and less often on MuSiQue.

Table 5.1: Inference retrieval count audit against DeepRAG on the overlapping multi-hop datasets.

Dataset	SEEK EM	SEEK retrievals/q	DeepRAG retrievals/q	Difference
HotpotQA	34.3	2.078	1.716	+21%
2Wiki	49.9	2.341	1.451	+61%
MuSiQue	24.8	1.840	2.273	-19%

The efficiency claim that survives this audit concerns offline adaptation and routing training, not guaranteed lower inference retrieval. SEEK trains only a LoRA decomposer; its retrieval gate, direct-vs-decompose verifier, and type-aware fallback are training-free. DeepRAG instead learns a retrieval policy through trajectory construction, SFT, and DPO-style preference optimisation.

The gradient-training FLOPs are of the same order, so the claim is not that SEEK is an order of magnitude cheaper in raw backpropagation. What SEEK saves is the DPO stage, the tree-search trajectory construction pipeline, and any learned retrieval-policy retraining.

Table 5.2: Offline adaptation cost comparison between SEEK and DeepRAG. DeepRAG training wall-clock was not reproduced locally; its FLOPs are an order estimate from the published training pipeline.

Component	SEEK	DeepRAG
Trained component	KG-triple decomposer only	Retrieval policy model
Fine-tuning type	LoRA SFT, less than 1% trainable parameters	Full-parameter SFT plus full-parameter DPO
Training stages	One SFT stage	Two stages: SFT and 8-epoch DPO
Measured wall-clock	37.9 minutes on one GPU	Not reproduced locally; expected multi-GPU hours
Gradient FLOPs	2.40×10^{17} measured	About 3.3×10^{17} estimated
Data construction	Direct supervised decompositions	Binary-tree-search trajectories; 72B follow-up model plus up to 8 7B vLLM servers in the official construction pipeline
Dataset adaptation	Reuse the same training-free routing rules	Rebuild trajectories and retrain the retrieval policy for new settings

This is the practical meaning of the "lightweight" claim in this thesis.

5.2 Why Direct and Decomposition Paths Must Coexist

The answer-level 2x2 analysis shows that decomposition is valuable but not always dominant. On 2Wiki, MuSiQue, and PopQA, the decompose-only column is much larger than the direct-only column, which confirms that decomposition solves many examples the direct path misses. HotpotQA is the exception: direct-only and decompose-only are nearly tied. This means HotpotQA is not simply an always-decompose dataset; the original question often carries comparison targets or disambiguating clues that are weakened by decomposition.

This is why SEEK keeps both paths. The direct path preserves the full question context, while the decomposition path serves as a retrieval strategy that exposes intermediate entities. Each path fails on different examples, and that is what makes the verifier necessary.

The case-level comparison with DeepRAG in Section 4.5.4 extends this point across systems. DeepRAG generates each follow-up question conditioned on the chain answered so far. This lets it adapt the plan during reasoning, but it also means a wrong intermediate answer can rewrite the rest of the plan: in the traces, follow-up questions issued after a wrong hop contain fragments of the earlier erroneous reasoning, and the chain keeps elaborating the wrong entity instead of recovering. SEEK's decomposition is produced up front from the original question alone. A wrong bridge answer still degrades the second-hop retrieval query, as the MuSiQue analysis in Section 5.3 shows, but it cannot corrupt the structure of the remaining sub-questions, and the direct path remains available as a fallback.

The retrieval decision differs in the same way. DeepRAG learns a retrieve-or-answer policy for each step, and on 2Wiki this policy becomes a liability: in 55 of the 160 questions that SEEK wins, DeepRAG retrieved nothing at all and answered the entire chain from parametric memory. SEEK's tail-leakage gate keeps step-level retrieval near or above 90% on the multi-hop datasets, so a fully parametric chain is rare. The price, documented in Table 5.1, is that SEEK retrieves more often overall; the benefit is that it seldom hallucinates a whole reasoning chain.

Original-question evidence union. Original-question evidence union is the most important methodological update in the final system. Decomposition improves answer-containing retrieval, but short sub-questions can lose the global context needed to interpret the relation correctly. By adding original-question evidence to each decomposed step, SEEK recovers part of this lost context without changing the decomposer or training a new retriever.

The cleanest comparison is within the decomposition path, because union changes only the decomposed sub-question context. Under this path-local ablation, HotpotQA's decomposed answer candidate improves from 25.00 to 30.50 EM, and 2Wiki improves from 47.20 to 50.20 EM. The full-system score also rises before the type-aware fallback, from 31.80 to 34.30 EM on HotpotQA and from 46.90 to 49.90 EM on 2Wiki, but that number additionally reflects the verifier's final path choices. This supports the thesis motivation directly: decomposition should make retrieval sharper, but it should not force

the system to forget the original question.

5.3 What Training the Decomposer Buys

The MuSiQue gold-decomposition analysis clarifies the role of decomposer training. SEEK's predicted decomposition raises answer-containing hit@5 from 19.0 to 38.8, while a gold decomposition with a gold intermediate answer reaches 55.8. When the SEEK sub-question text is given the gold bridge entity, it reaches 54.1, nearly the same as gold. This means the decomposer's wording is not the main bottleneck; the main bottleneck is whether the first-hop bridge entity is answered correctly before the second query is formed.

The untuned base-model comparison gives the other half of the story. The base Llama-3-8B-Instruct model can produce useful query text under oracle substitution, but it produces a valid two-step decomposition for only 149 of 1,000 MuSiQue examples and has a 15.3% parsing failure rate. LoRA training therefore buys structural reliability: valid step counts, valid placeholders, and compact query style. That reliability is what makes the decomposition path executable at inference time.

5.4 Tail-Leakage Surprisal: Useful Signal, Not Correctness Guarantee

Tail-leakage surprisal remains useful for deciding whether a sub-question should retrieve. The never-retrieve, gate, and always-retrieve ablation shows that the gate closely matches always-retrieve accuracy while saving a modest amount of retrieval. The gate-signal replay and the direct-path confidence battery also show that tail-leakage surprisal is the strongest of the tested uncertainty signals on average, ahead of maximum entropy and the other alternatives.

However, it is not a correctness guarantee. The direct-path analysis shows a stable confident-wrong region: 7% to 22% of wrong direct answers carry surprisal at or below the median of correct answers, and the verifier-side quadrant analysis paints the same picture for query-level entropy.

A compact way to state these findings is that confidence signals measure properties of the answer distribution, not the correctness of the reasoning behind it. The most common confident error in our traces is an incomplete reasoning path: the model stops at the first-hop bridge entity, or returns an attribute where the question asks for a comparison winner, and it does so with the same near-zero leakage as a fully correct answer. In a separate diagnostic experiment that sampled each question fifteen times without retrieval, 1,538 of 4,000 questions produced fifteen identical answers and hence zero dispersion, yet 40.3% of those answers were wrong. No amount of resampling can flag such cases, because there is no disagreement to detect, and the confident-wrong analysis in Chapter 4 shows the same failure for tail-leakage surprisal: clean parametric hallucinations are produced with essentially zero leakage. The mirror group of low-confidence correct answers also deserves care: nearly half of them pass only through token-F1 overlap rather than exact match, so they are officially correct but not always semantically correct, and high surprisal is an equally poor reason to discard an answer.

Confidence is also of little use for choosing between the two paths. When the direct and decomposed answers disagree, picking the path with the lower uncertainty performs no better than chance once the two scales are normalised, with per-dataset accuracy between 45% and 69%. The verifier's advantage on these disagreements comes from its structural rules and its decomposition prior, not from comparing confidence values.

This explains why the final system uses confidence inside a broader verifier rather than as a single hard router. Tail-leakage surprisal locates uncertainty at the step level, but the direct-vs-decompose choice also needs answer type, decomposition stability, and candidate agreement. Even then, the remaining confidently wrong cases point toward future evidence-checking or self-consistency methods rather than another global threshold.

5.5 Remaining Headroom and Remaining Problems

The oracle analysis gives a measured upper bound for better routing. In the final traces, oracle direct-vs-decompose routing reaches 41.60 EM on HotpotQA, 54.10 on 2Wiki, 28.60 on MuSiQue, and 55.70 on PopQA. The current verifier still leaves 3.8 to 6.1 EM of routing headroom.

This headroom should be interpreted carefully. Oracle routing is not deployable because it uses the gold answer after the fact. It also cannot fix examples where both paths are wrong. The final traces contain 584 both-wrong HotpotQA examples, 459 2Wiki examples, 714 MuSiQue examples, and 443 PopQA examples. These examples require improvements below the router: better retrieval, reranking, bridge-answer propagation, or final synthesis.

The failure taxonomy makes the bottleneck concrete. Among decomposition-path errors, retrieval gaps account for 47% of HotpotQA errors, 64% of 2Wiki errors, and 68% of MuSiQue errors. Synthesis failures contribute another 15% to 18%. Constraint loss is smaller overall but important on HotpotQA, where it explains why original-question evidence union gives the largest gain. Thus the remaining problem after SEEK is not one problem; it splits into routing headroom, retrieval coverage, synthesis robustness, and confident wrong answers.

5.6 Threats to Validity

External baseline numbers are paper numbers. The external baseline results in Table 4.1 are taken from the DeepRAG paper rather than reproduced in the exact same harness. Normalisation, prompting, and implementation details may differ slightly. To reduce protocol mismatch, SEEK uses the honest wiki-21M pool and the same first-1,000-question evaluation style, and the reported averages are recomputed only over the four overlapping datasets.

Dataset slicing. All headline results use the first 1,000 dev questions to align with DeepRAG. MuSiQue dev is sorted by hop depth, so this slice understates full-dev difficulty. Full-dev MuSiQue remains harder, especially for 4-hop questions.

Single seed and greedy decoding. All numbers use greedy decoding without seed averaging. Small differences of 1 to 2 EM should be interpreted cautiously; larger effects such as PopQA's gain over DeepRAG, the 2Wiki improvement, and the original-question evidence union gains are more robust.

Routing cannot fix both-wrong cases. The verifier can only choose between two existing candidates. If both direct and decomposed answers are wrong, routing cannot recover the example. This is the main reason oracle routing is an upper bound on routing quality rather than an upper bound on the whole QA system.

Chapter 6 Conclusions and Future Work

6.1 Summary of Contributions

This thesis introduced **SEEK**, a lightweight adaptive RAG framework for open-domain question answering. The main contribution is to frame adaptive RAG as a *decomposition-routing problem*: the system must decide not only when to retrieve, but also when to trust a decomposed reasoning path and when to preserve the original question.

The principal contributions are:

1. **Dual-path adaptive RAG.** SEEK generates both a direct answer and a decomposed answer, allowing the final system to compare original-question reasoning with step-by-step decomposition.

2. **KG-triple decomposition for sub-question retrieval.** The decomposer emits compact `entity >> relation` fragments with `#N` dependencies, making multi-hop retrieval queries explicit and executable.

3. **Surprisal-gated sub-question retrieval.** Tail-leakage surprisal, the summed surprisal of the probability mass escaping the model's own top-five candidates, provides a transparent and training-free signal for deciding whether each decomposed step should retrieve external evidence.

4. **Original-question evidence preservation.** The final system unions original-question evidence into each decomposed step, reducing context loss caused by overly short sub-questions.

5. **Training-free verification.** SEEK selects between direct and decomposed candidates using confidence, answer type, decomposition stability, and candidate agreement, without classifier training, tree search, DPO, or preference data.

6. **Honest wiki-21M evaluation.** Under the DeepRAG-aligned open-domain pool, SEEK reaches 41.62 average EM and 48.66 average F1, outperforming DeepRAG on average and winning on 2Wiki, MuSiQue, and PopQA.

6.2 Limitations

The principal limitations are:

- **HotpotQA remains below DeepRAG.** SEEK improves HotpotQA to 37.50 EM, but still trails DeepRAG's 40.70 EM. HotpotQA is the clearest case where direct and decomposed paths are complementary but the current verifier does not fully exploit the headroom.
- **Verifier headroom remains.** Oracle direct-vs-decompose routing still offers 3.8 to 6.1 EM over the current verifier, so routing is not solved.
- **Confidence is not a correctness guarantee.** Tail-leakage surprisal is useful for sub-question retrieval gating, but 7% to 22% of wrong direct answers look as confident as typical correct answers. A battery of more than forty alternative confidence signals did not find a decisive per-question error detector either (Section 4.6).
- **Bridge-answer drift limits decomposition.** On MuSiQue, SEEK nearly matches gold decomposition when the first-hop bridge entity is correct, but wrong intermediate answers drag the second query away from the target evidence.
- **Both-wrong cases require better retrieval and synthesis.** Routing cannot fix examples where both direct and decomposed candidates are wrong; retrieval gaps explain 47% to 68% of decomposition-path errors on the multi-hop datasets.
- **Single-seed greedy decoding.** Variance estimates are not reported and would strengthen small-difference claims.

6.3 Future Work

Stronger verifier signals. The largest immediate opportunity is improving the verifier. Candidate signals include self-consistency disagreement, top-1 versus top-2 logit margin, hidden-state variance, answer-type compatibility, and lightweight learned calibration from a small validation set. The signal battery in Chapter 4 suggests that gains from any single scalar signal are modest, so combining confidence with evidence-side checks is more

promising than searching for another scalar. The goal is to approach the oracle direct-vs-decompose headroom without losing SEEK's interpretability.

Better retrieval and reranking. The large both-wrong counts show that routing alone is insufficient. Dense retrieval, hybrid BM25+dense retrieval, or reranking over the unioned evidence could improve cases where neither path retrieves enough supporting information.

Length-conditioned decomposer. A target-depth control token such as `[K=1]`, `[K=2]`, or `[K=4]` could reduce negative transfer between single-hop and multi-hop data. The verifier or a small complexity predictor could choose the target depth at inference time.

Multi-sample decomposition. Sampling multiple decompositions and selecting by final-answer confidence or agreement could reduce wrong-first-hop propagation. This is especially relevant because the MuSiQue analysis shows that bridge-answer drift, rather than decomposition wording alone, explains much of the gap between SEEK and gold decomposition. This would trade inference cost for robustness without requiring DPO.

Evidence-aware verification. Confidently wrong answers cannot be removed by another confidence threshold alone. Future verifiers should test whether the selected answer is externally supported by retrieved evidence, possibly with lightweight entailment checks, answer-type validation, or self-consistency over evidence snippets.

Span calibration. A small post-training or post-processing module for terse answer spans could close EM-only failures while preserving F1. This is especially relevant for verbose entity descriptions and comparison questions.

6.4 Closing Remarks

This thesis argues that adaptive RAG should ask two questions: when should the system retrieve, and when should it trust decomposition? SEEK shows that a lightweight system can answer both questions with interpretable inference-time signals. Decomposition improves retrieval, tail-leakage surprisal helps control sub-question evidence use,

original-question evidence preserves global context, and a training-free verifier makes the final route explicit. The remaining oracle gap leaves a clear path for future work, but the central result is already useful: adaptive RAG can be competitive with heavier learned-policy systems without hiding its routing decisions inside another trained model.

References

[1] Y. Gao et al., “Retrieval-augmented generation for large language models: A survey”, *arXiv preprint arXiv:2312.10997*, 2023.

[2] L. Huang et al., “A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions”, *ACM Transactions on Information Systems*, 2025.

[3] Y. Huang and J. Huang, “A survey on retrieval-augmented text generation for large language models”, *arXiv preprint arXiv:2404.10981*, 2024.

[4] A. Mallen, A. Asai, V. Zhong, R. Das, D. Khashabi, and H. Hajishirzi, “When not to trust language models: Investigating effectiveness of parametric and non-parametric memories”, in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

[5] R. Ren et al., “Investigating the factual knowledge boundary of large language models with retrieval augmentation”, *arXiv preprint arXiv:2307.11019*, 2023.

[6] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, “Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions”, in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

[7] S. Jeong, J. Baek, S. Cho, S. J. Hwang, and J. C. Park, “Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity”, in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2024.

[8] Y. Wang, P. Li, M. Sun, and Y. Liu, “Self-knowledge guided retrieval augmentation for large language models”, in *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023.

[9] Z. Jiang et al., “Active retrieval augmented generation”, in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[10] W. Su, Y. Tang, Q. Ai, Z. Wu, and Y. Liu, “Dragin: Dynamic retrieval augmented generation based on the real-time information needs of large language models”, in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024.

[11] Z. Yao et al., “Seakr: Self-aware knowledge retrieval for adaptive retrieval augmented generation”, in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025.

[12] Y. Wang, Lu wei, and H. Ling, “Retrieval as a decision: Training-free adaptive gating for efficient rag”, *arXiv preprint arXiv:2511.09803*, 2025.

[13] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, “Self-rag: Learning to retrieve, generate, and critique through self-reflection”, in *International Conference on Learning Representations (ICLR)*, 2024.

[14] X. Guan et al., “Deeprag: Thinking to retrieve step by step for large language models”, in *International Conference on Learning Representations (ICLR)*, 2026.

[15] Su et al., “Fast or better? balancing accuracy and cost in retrieval-augmented generation with flexible user control”, *CoRR*, 2025.

[16] S. Min, V. Zhong, L. Zettlemoyer, and H. Hajishirzi, “Multi-hop reading comprehension through question decomposition and rescoring”, in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 6097–6109.

[17] T. Wolfson et al., “Break it down: A question understanding benchmark”, *Transactions of the Association for Computational Linguistics*, vol. 8, pp. 183–198, 2020.

[18] T. Yu, S. Zhang, and Y. Feng, “Auto-rag: Autonomous retrieval-augmented generation for large language models”, *arXiv preprint arXiv:2411.19443*, 2024.

[19] X. Li et al., “Search-o1: Agentic search-enhanced large reasoning models”, *arXiv preprint arXiv:2501.05366*, 2025.

[20] W. Feng, C. Hao, Y. Zhang, J. Song, and H. Wang, “Airrag: Activating intrinsic reasoning for retrieval-augmented generation via tree-based search”, *arXiv preprint arXiv:2501.10053*, 2025.

[21] S. Kadavath et al., “Language models (mostly) know what they know”, *arXiv preprint arXiv:2207.05221*, 2022.

[22] Z. Yin, Q. Sun, Q. Guo, J. Wu, X. Qiu, and X. Huang, “Do large language models know what they don’t know?”, in *Findings of the Association for Computational Linguistics: ACL 2023*, 2023.

[23] S. Kapoor et al., “Large language models must be taught to know what they don’t know”, *arXiv preprint arXiv:2406.08391*, 2024.

[24] X. Yin, X. Zhang, J. Ruan, and X. Wan, “Benchmarking knowledge boundary for large language models: A different perspective on model evaluation”, *arXiv preprint arXiv:2402.11493*, 2024.

[25] Y. Yang, H. Yoo, and H. Lee, “Maqa: Evaluating uncertainty quantification in llms regarding data uncertainty”, in *Findings of the Association for Computational Linguistics: NAACL 2025*, 2025.

[26] L. Kuhn, Y. Gal, and S. Farquhar, “Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation”, in *International Conference on Learning Representations (ICLR)*, 2023.

[27] S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, “Detecting hallucinations in large language models using semantic entropy”, *Nature*, vol. 630, pp. 625–630, 2024.

[28] Z. Yang et al., “Hotpotqa: A dataset for diverse, explainable multi-hop question answering”, in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018.

[29] X. Ho, A.-K. Duong Nguyen, S. Sugawara, and A. Aizawa, “Constructing a multi-hop qa dataset for comprehensive evaluation of reasoning steps”, in *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*, 2020.

[30] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, “Musique: Multihop questions via single-hop question composition”, *Transactions of the Association for Computational Linguistics (TACL)*, 2022.

[31] T. Kwiatkowski et al., “Natural questions: A benchmark for question answering research”, *Transactions of the Association for Computational Linguistics (TACL)*, 2019.

[32] M. Joshi, E. Choi, D. S. Weld, and L. Zettlemoyer, “Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension”, in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2017.

[33] P. Rajpurkar, J. Zhang, K. Lopyrev, and P. Liang, “Squad: 100,000+ questions for machine comprehension of text”, in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2016.

[34] E. J. Hu et al., “Lora: Low-rank adaptation of large language models”, in *International Conference on Learning Representations (ICLR)*, 2022.

[35] Y. Zheng et al., “Llamafactory: Unified efficient fine-tuning of 100+ language models”, in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 3: System Demonstrations)*, 2024.

Appendices

Appendix A Supplementary Material

This appendix contains supplementary material such as full prompt templates, additional experimental results, dataset statistics, hyperparameter settings, and source code snippets.